

BFS Traversal

Step 1: Start at A

Legend



Processing

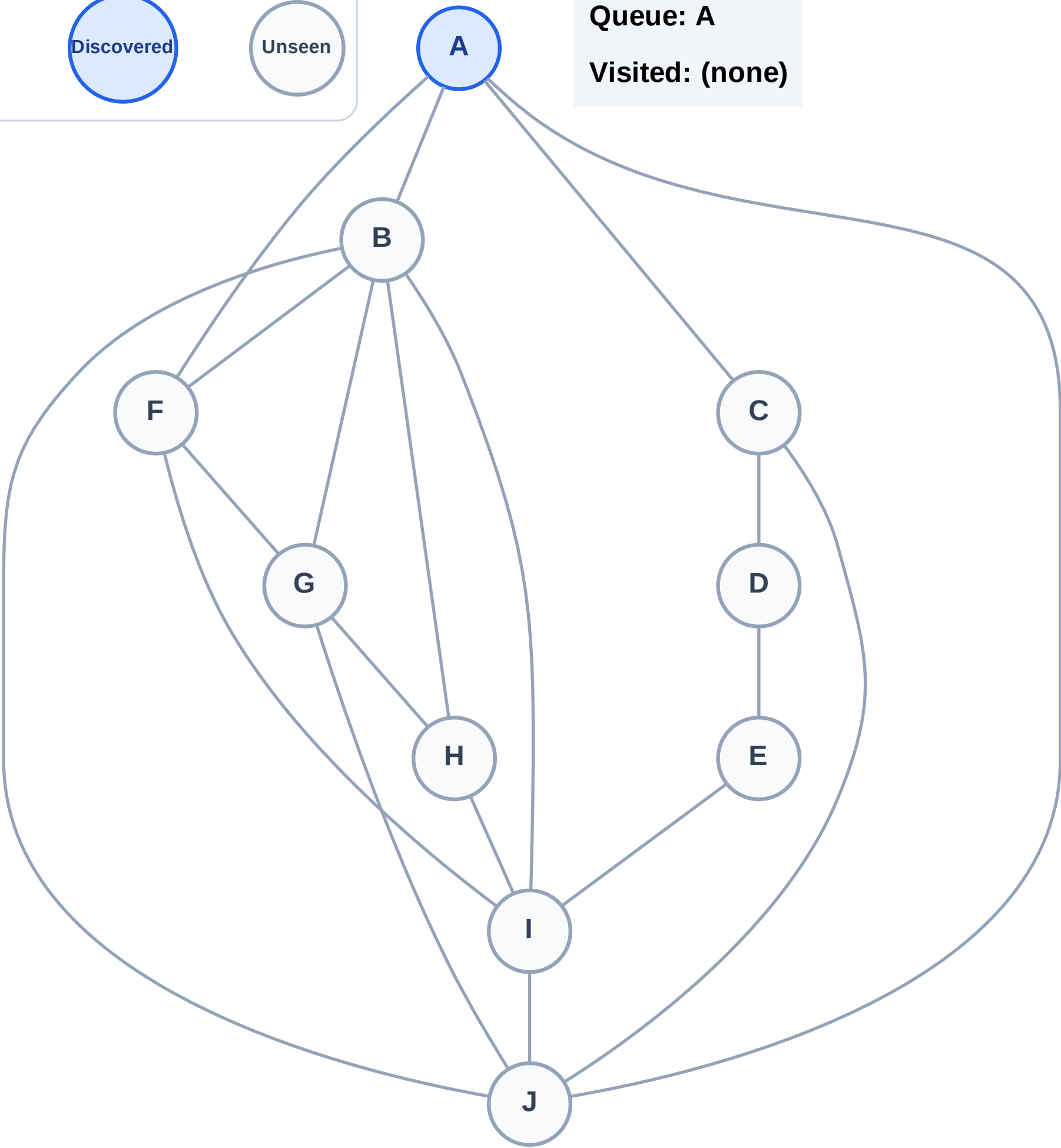


Unseen



Queue: A

Visited: (none)



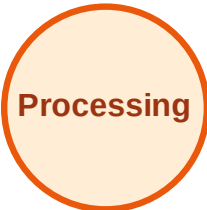
BFS Traversal

Step 2: Process node A

Legend



Processing



Discovered

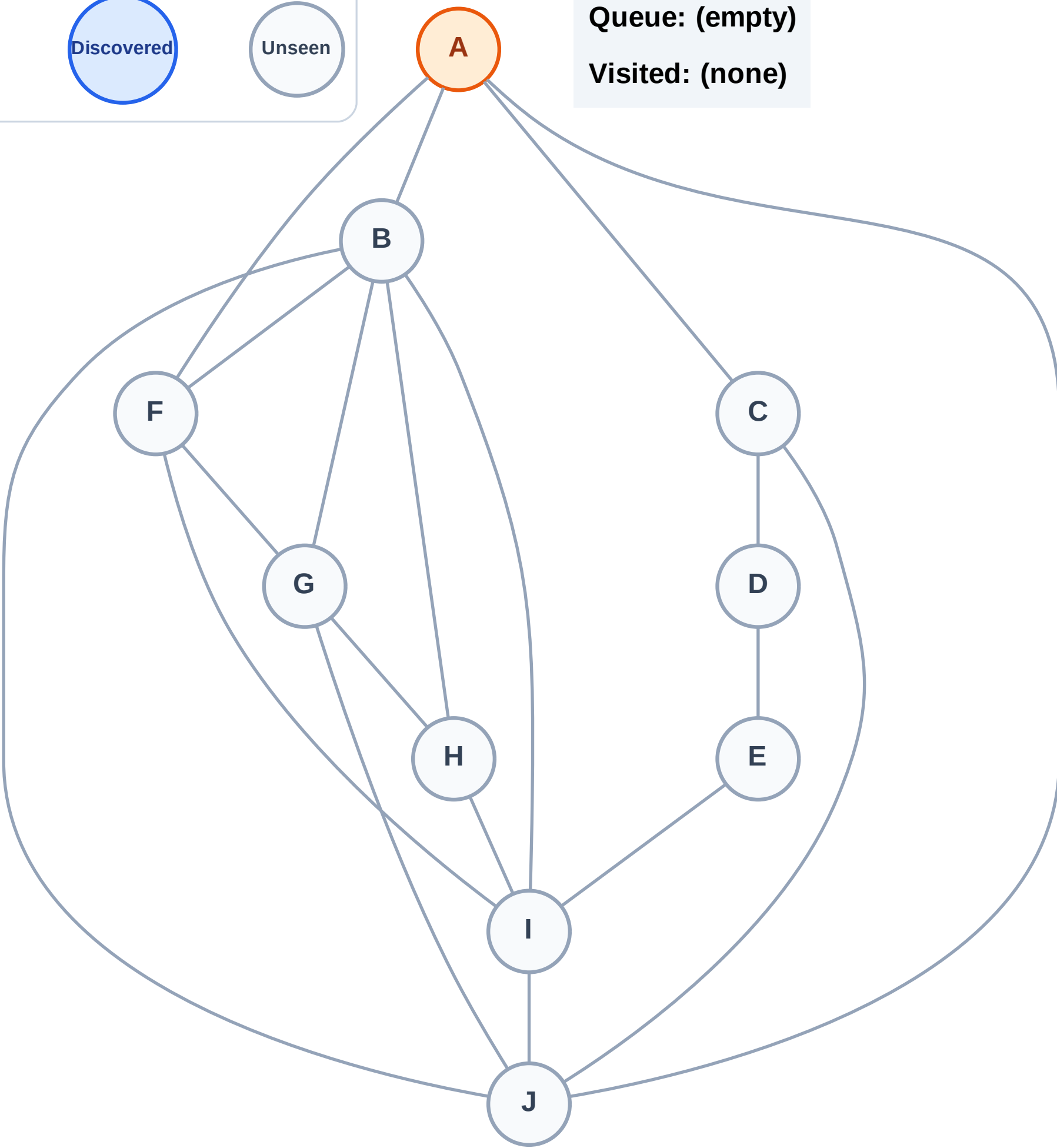


Unseen



Queue: (empty)

Visited: (none)



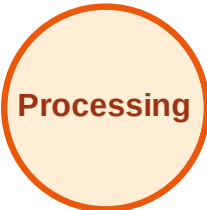
BFS Traversal

Step 3: Discover B, C, F, J from A

Legend



Complete



Processing



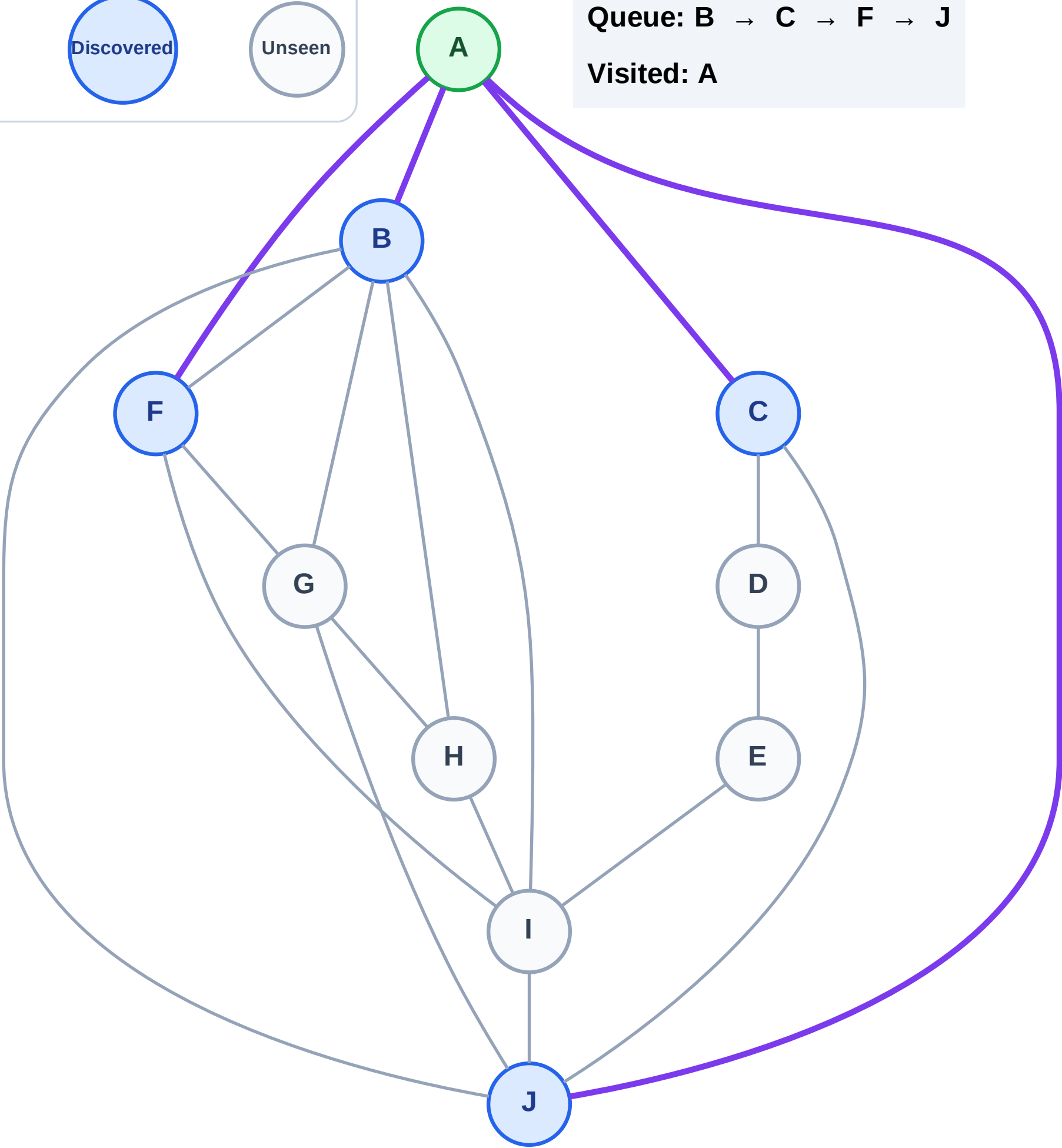
Discovered



Unseen

Queue: B → C → F → J

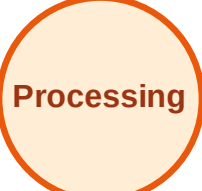
Visited: A



BFS Traversal

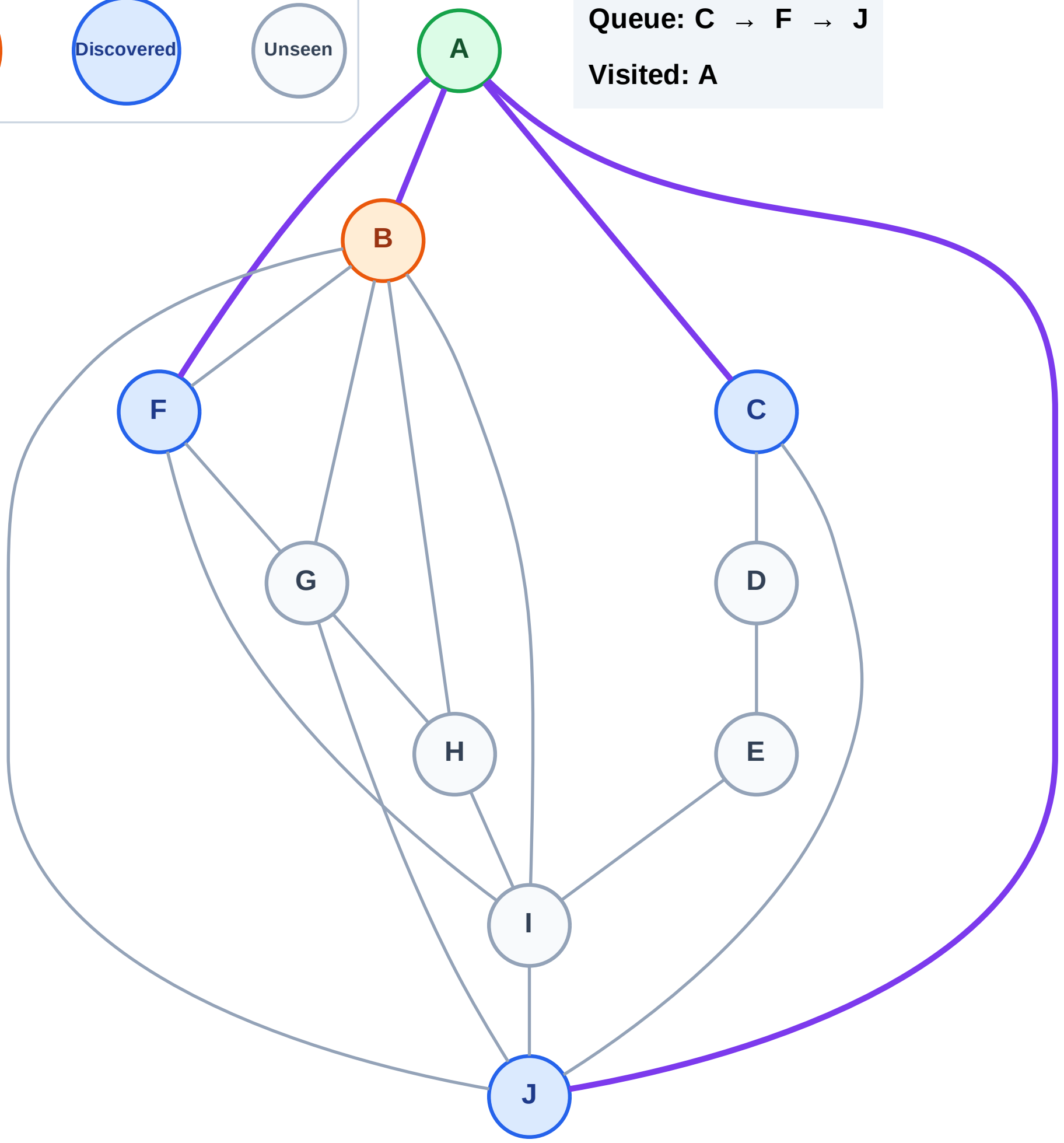
Step 4: Process node B

Legend



Queue: C → F → J

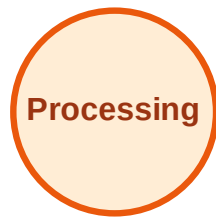
Visited: A



BFS Traversal

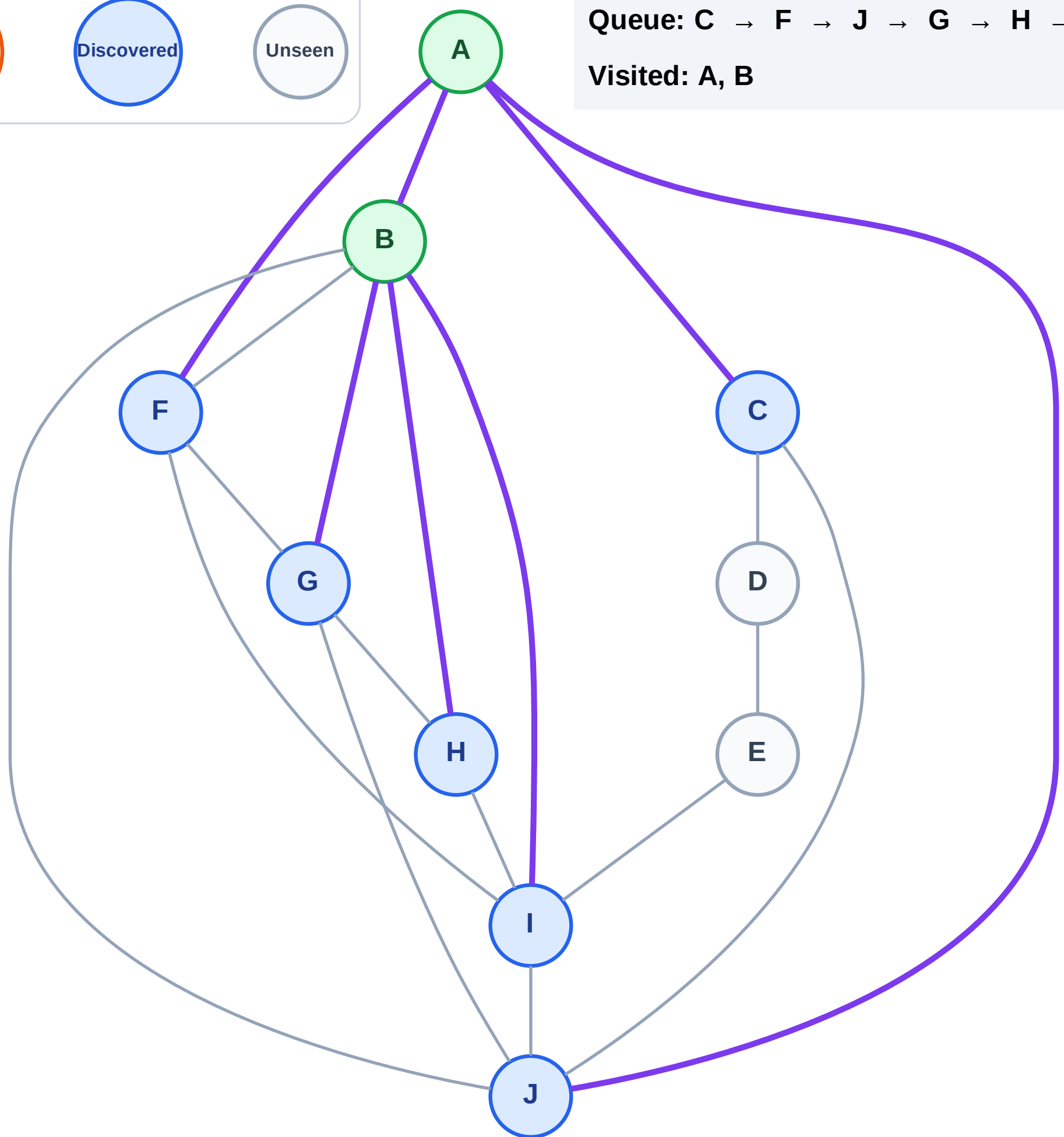
Step 5: Discover G, H, I from B

Legend



Queue: C → F → J → G → H → I

Visited: A, B



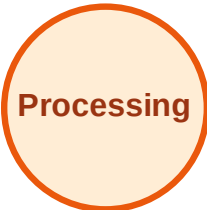
BFS Traversal

Step 6: Process node C

Legend



Complete



Processing



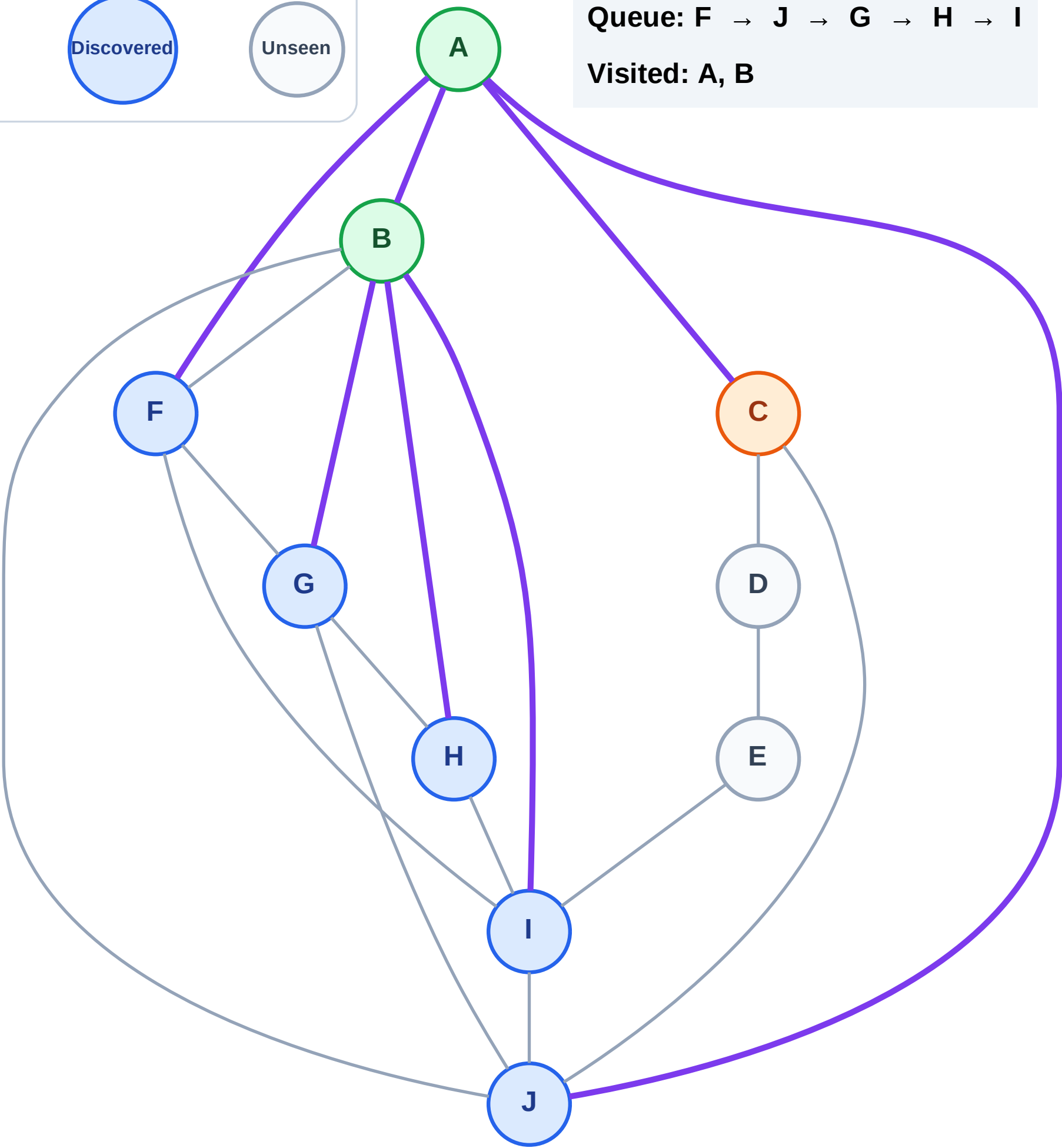
Discovered



Unseen

Queue: F → J → G → H → I

Visited: A, B



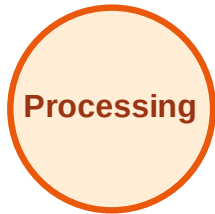
BFS Traversal

Step 7: Discover D from C

Legend



Complete



Processing



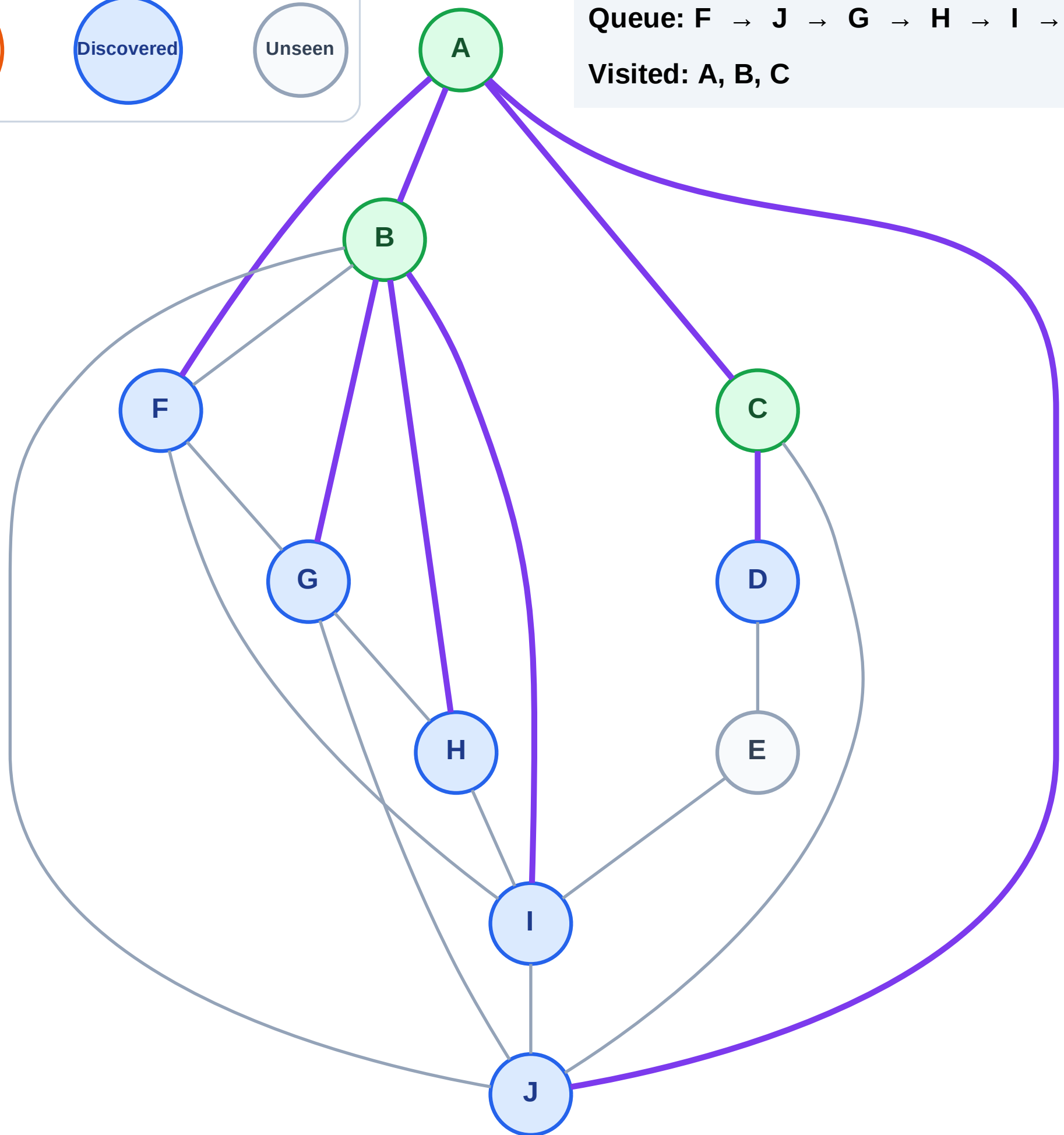
Discovered



Unseen

Queue: F → J → G → H → I → D

Visited: A, B, C



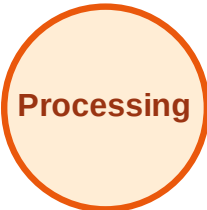
BFS Traversal

Step 8: Process node F

Legend



Complete



Processing



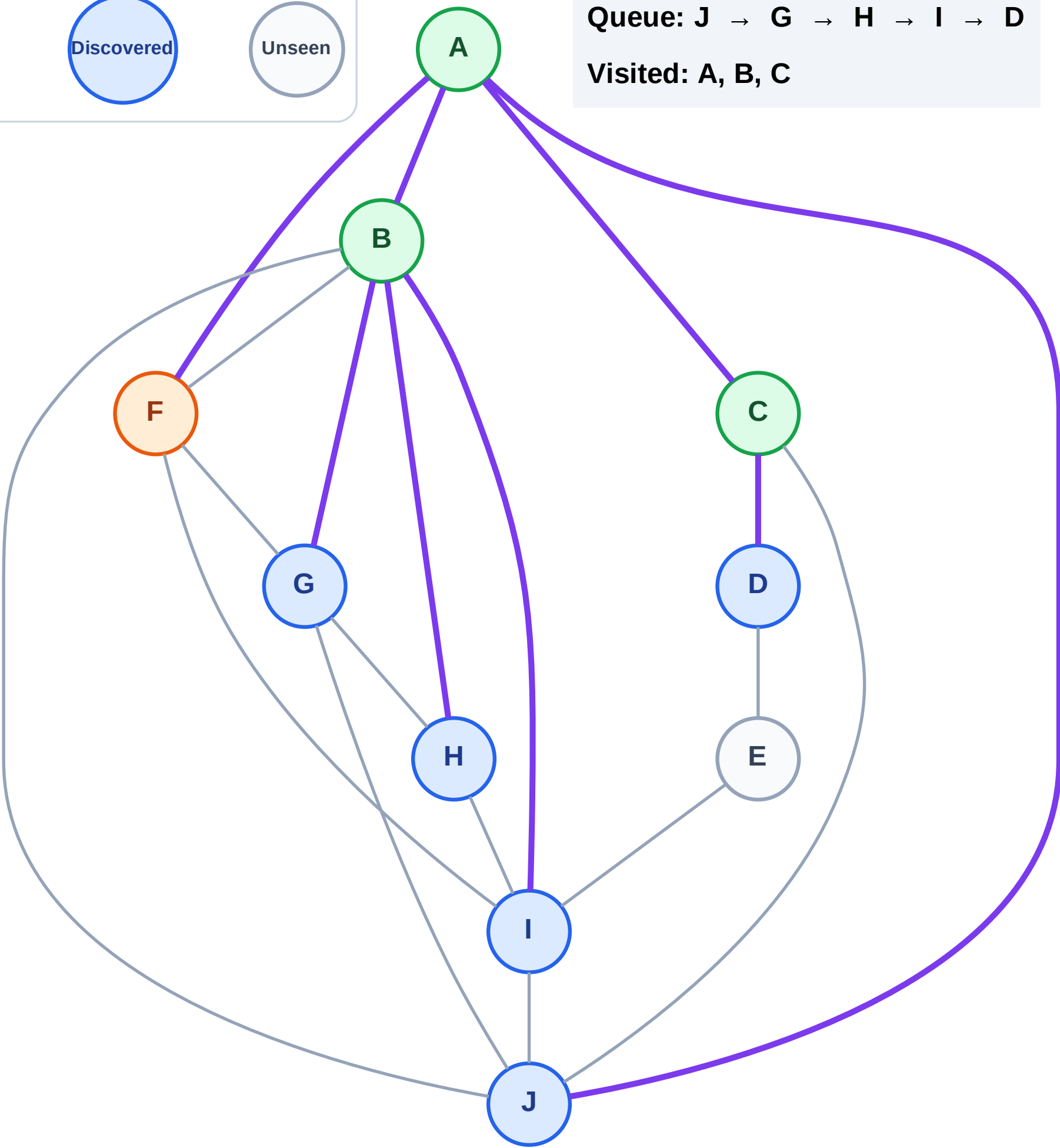
Discovered



Unseen

Queue: J → G → H → I → D

Visited: A, B, C



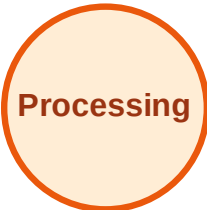
BFS Traversal

Step 9: No new neighbors from F

Legend



Complete



Processing



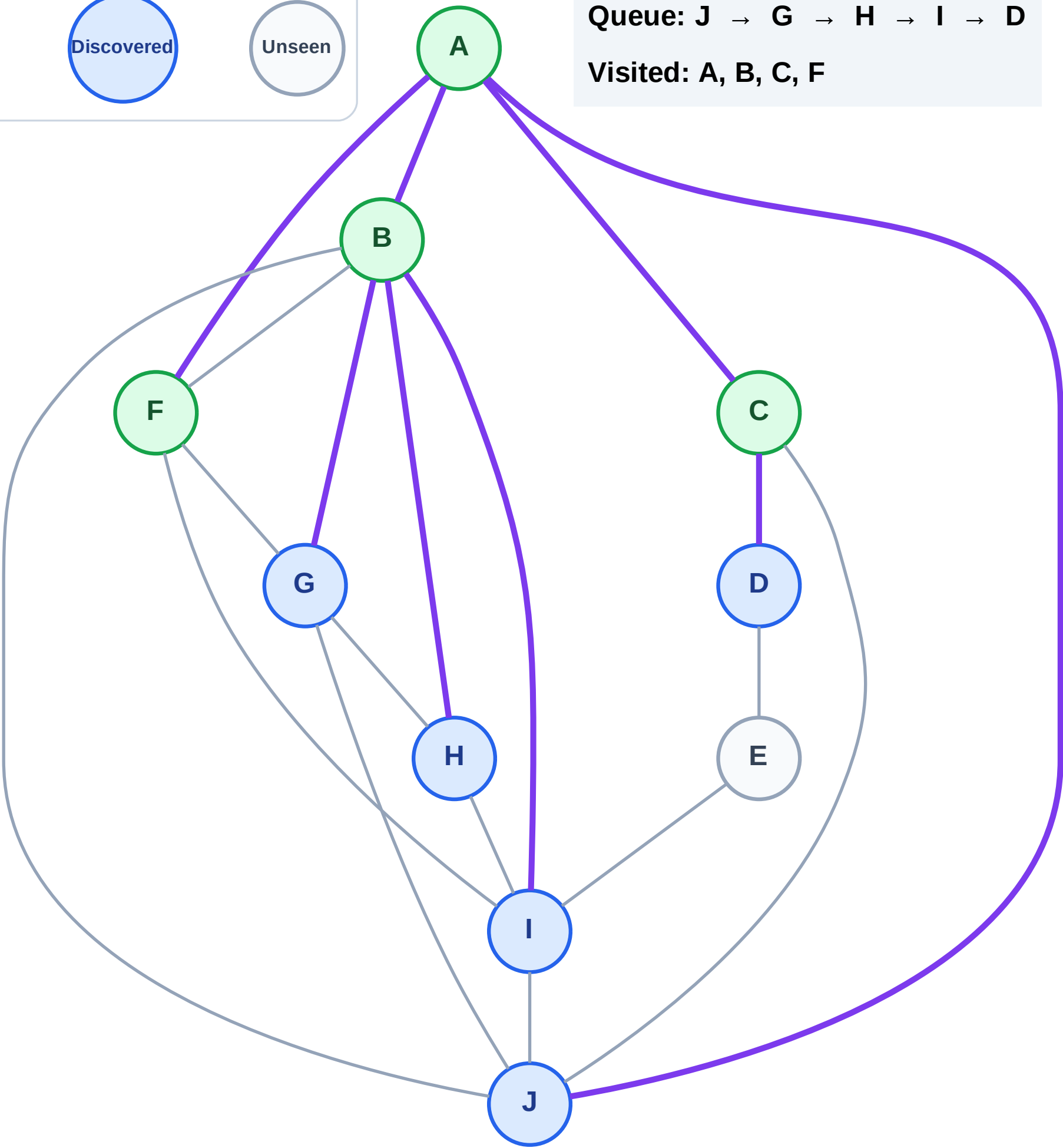
Discovered



Unseen

Queue: J → G → H → I → D

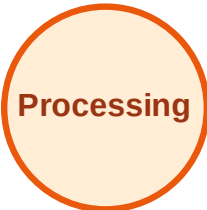
Visited: A, B, C, F



BFS Traversal

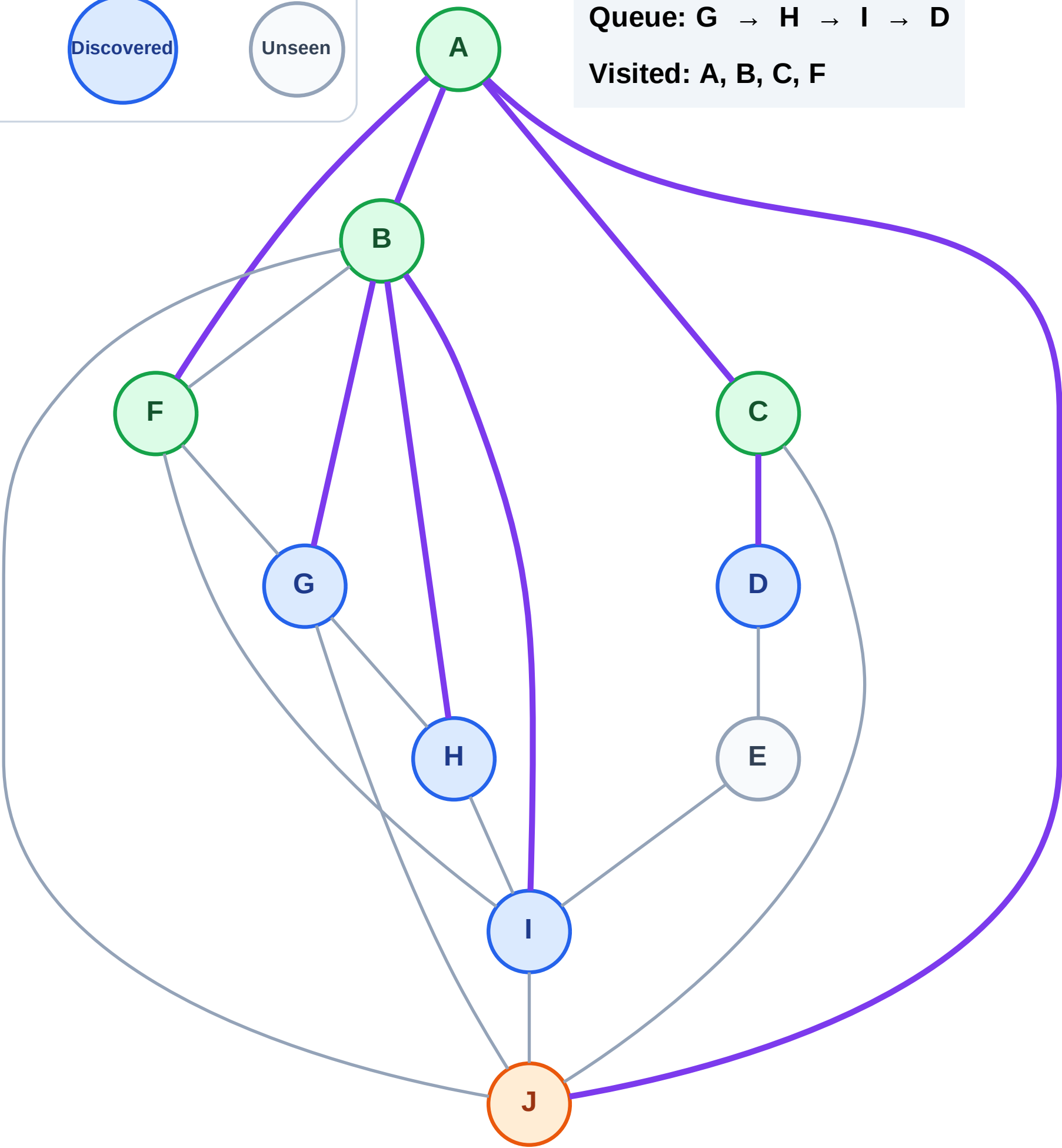
Step 10: Process node J

Legend



Queue: G → H → I → D

Visited: A, B, C, F



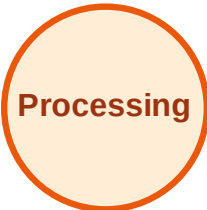
BFS Traversal

Step 11: No new neighbors from J

Legend



Complete



Processing



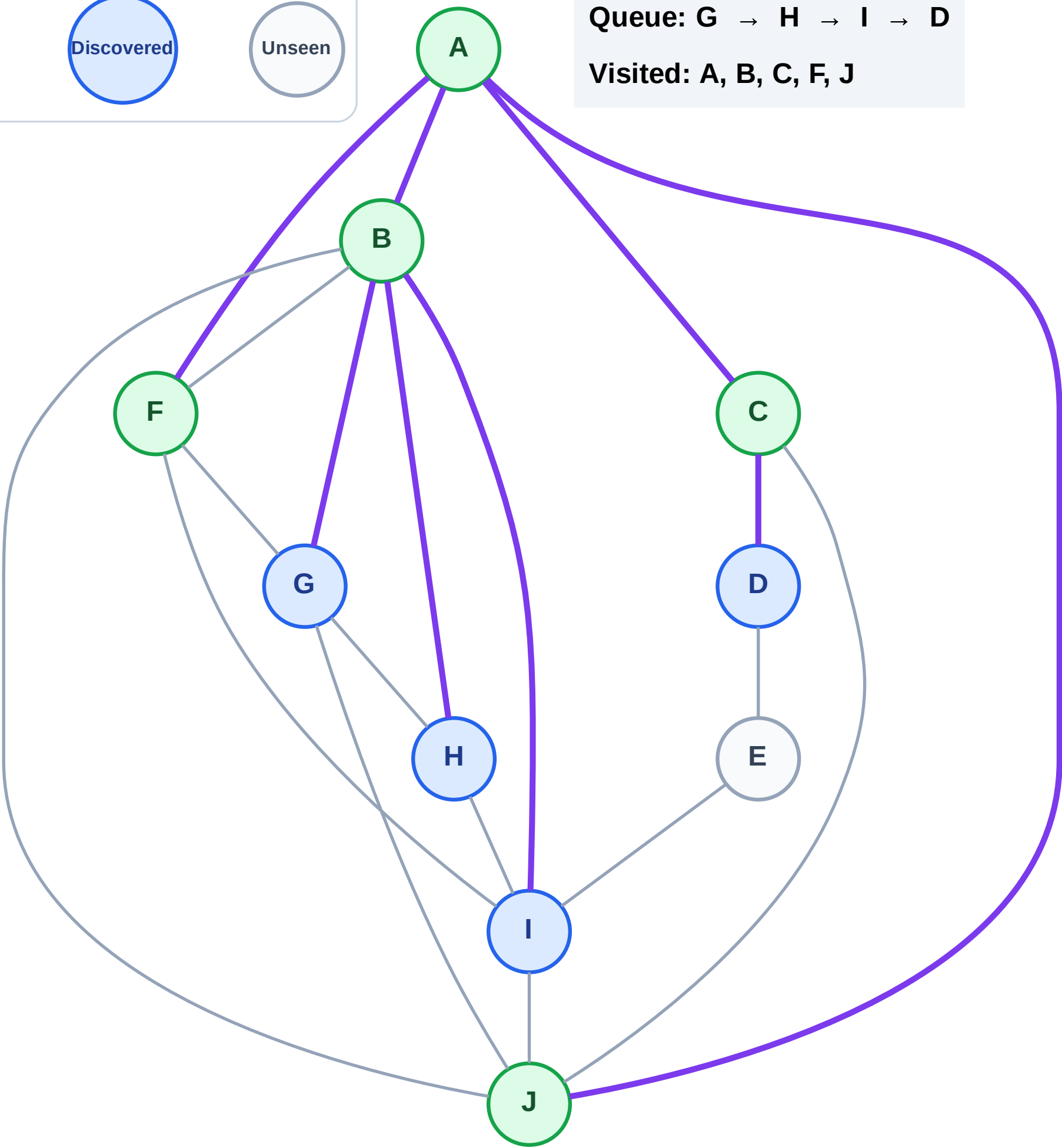
Discovered



Unseen

Queue: G → H → I → D

Visited: A, B, C, F, J



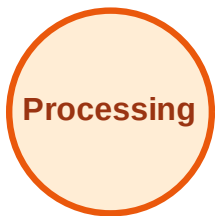
BFS Traversal

Step 12: Process node G

Legend



Complete



Processing



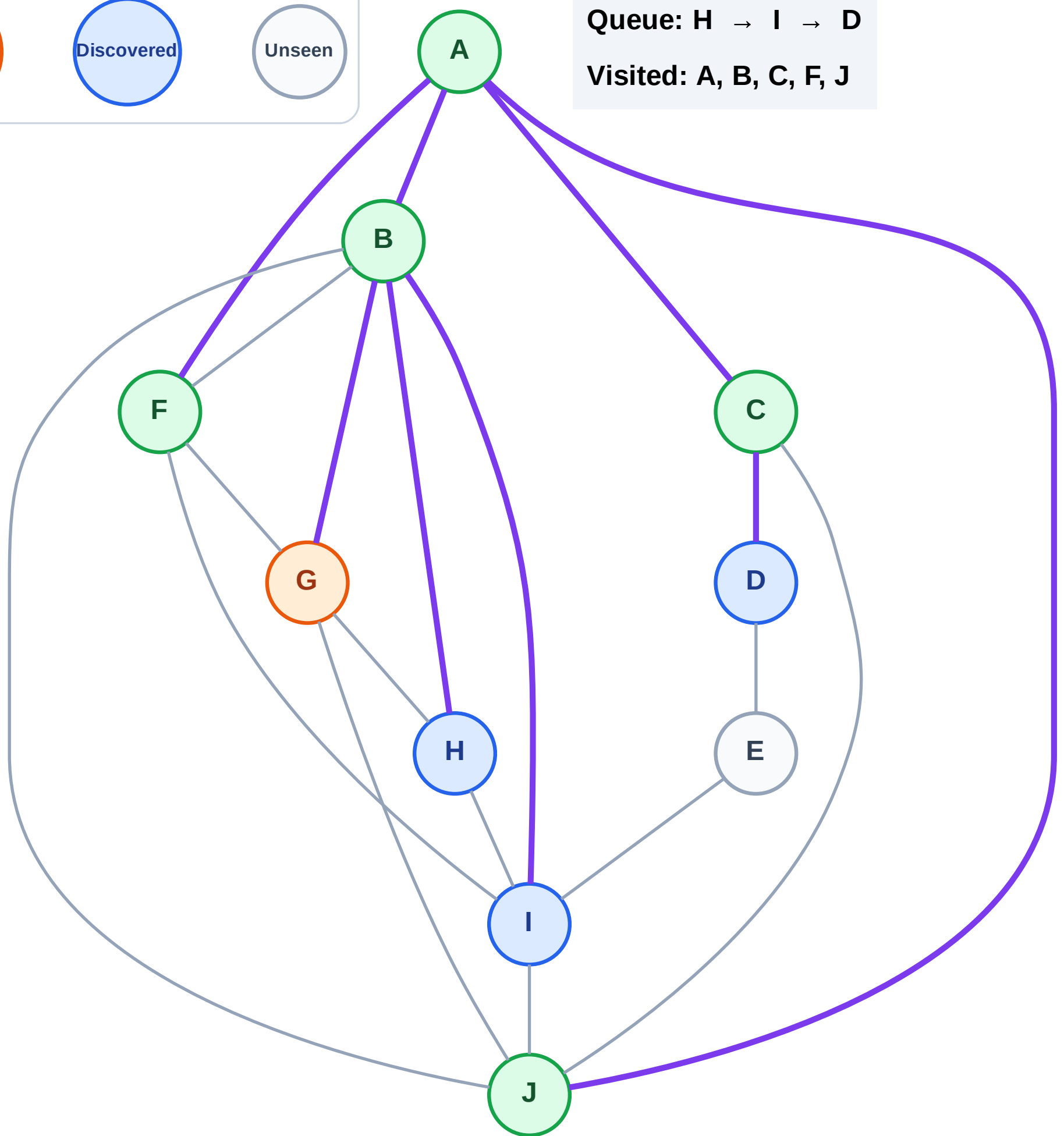
Discovered



Unseen

Queue: H → I → D

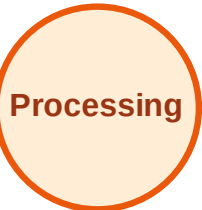
Visited: A, B, C, F, J



BFS Traversal

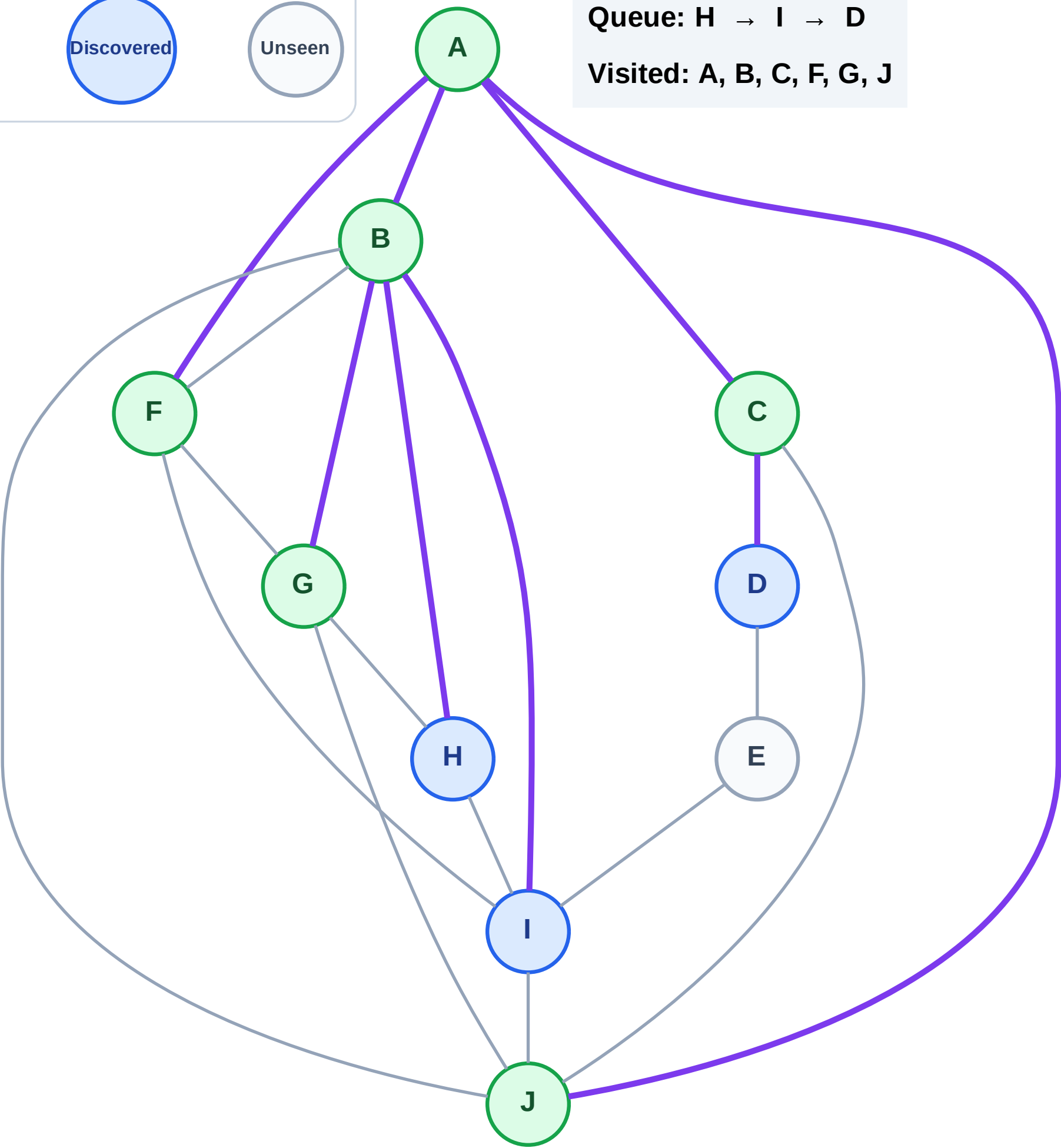
Step 13: No new neighbors from G

Legend



Queue: H → I → D

Visited: A, B, C, F, G, J



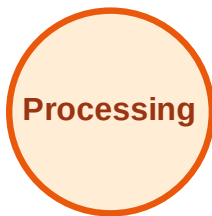
BFS Traversal

Step 14: Process node H

Legend



Complete



Processing



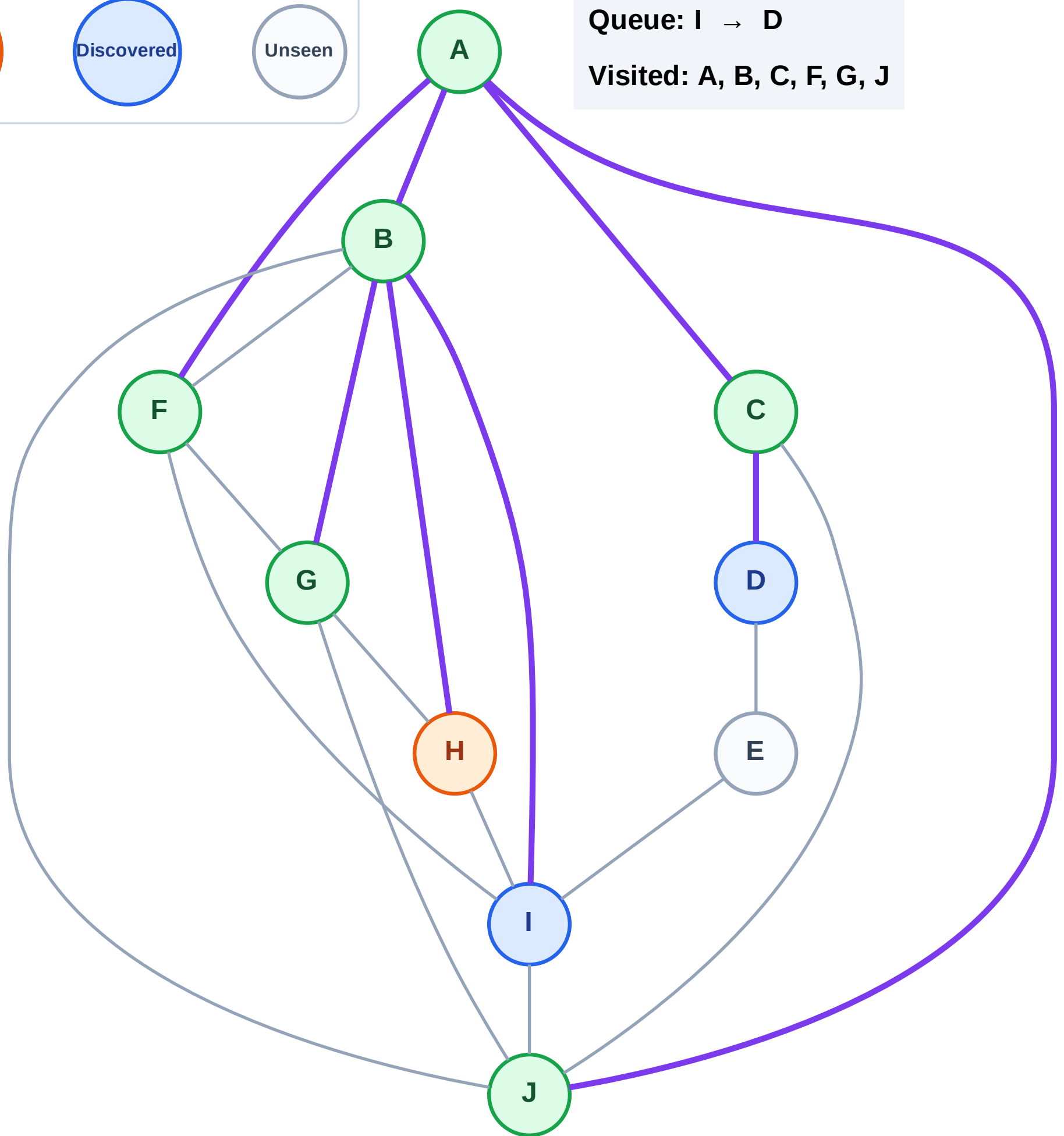
Discovered



Unseen

Queue: I → D

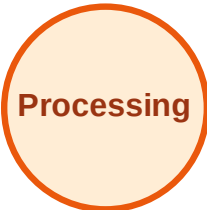
Visited: A, B, C, F, G, J



BFS Traversal

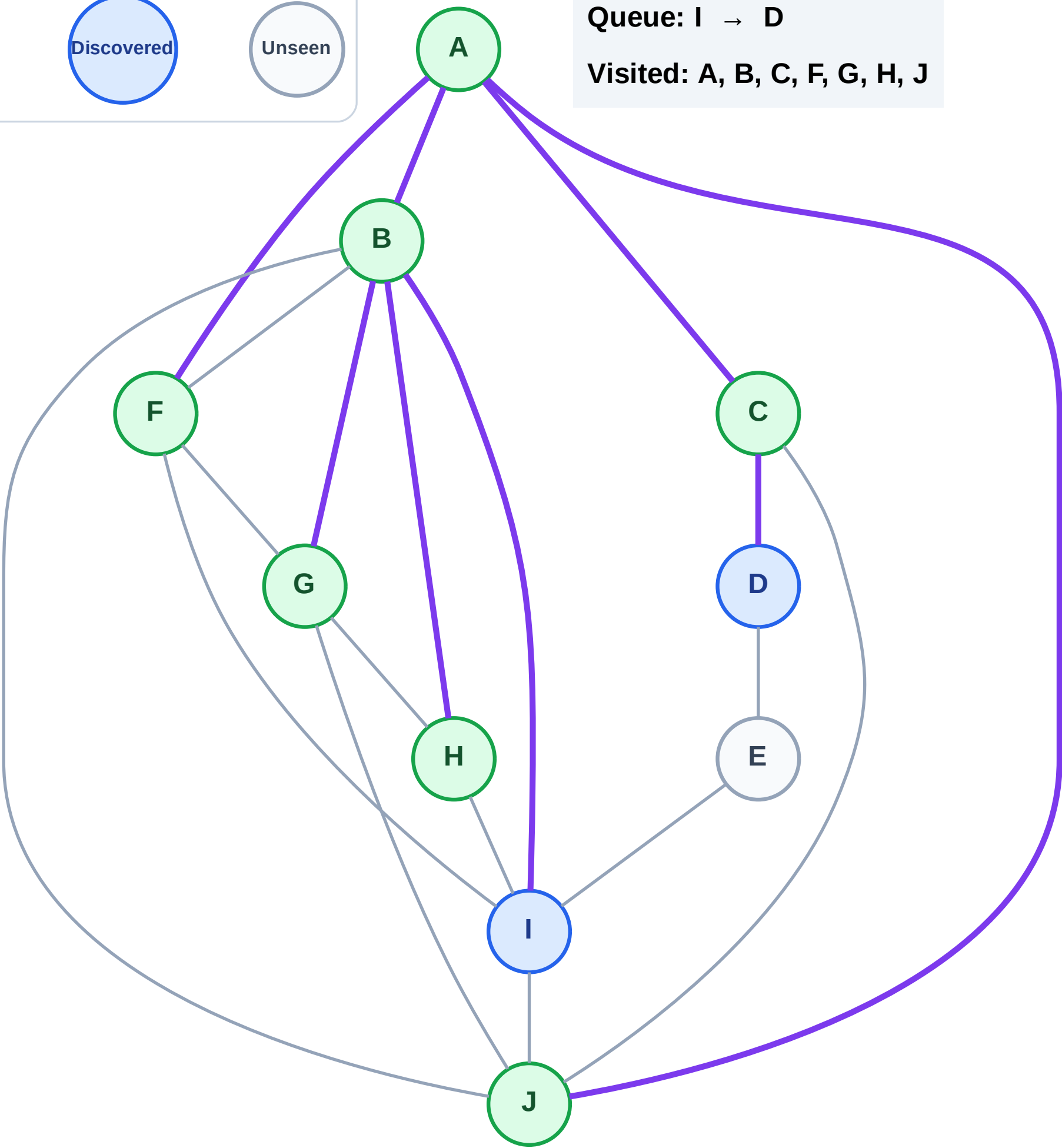
Step 15: No new neighbors from H

Legend



Queue: I → D

Visited: A, B, C, F, G, H, J



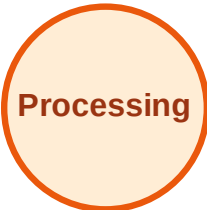
BFS Traversal

Step 16: Process node I

Legend



Complete



Processing



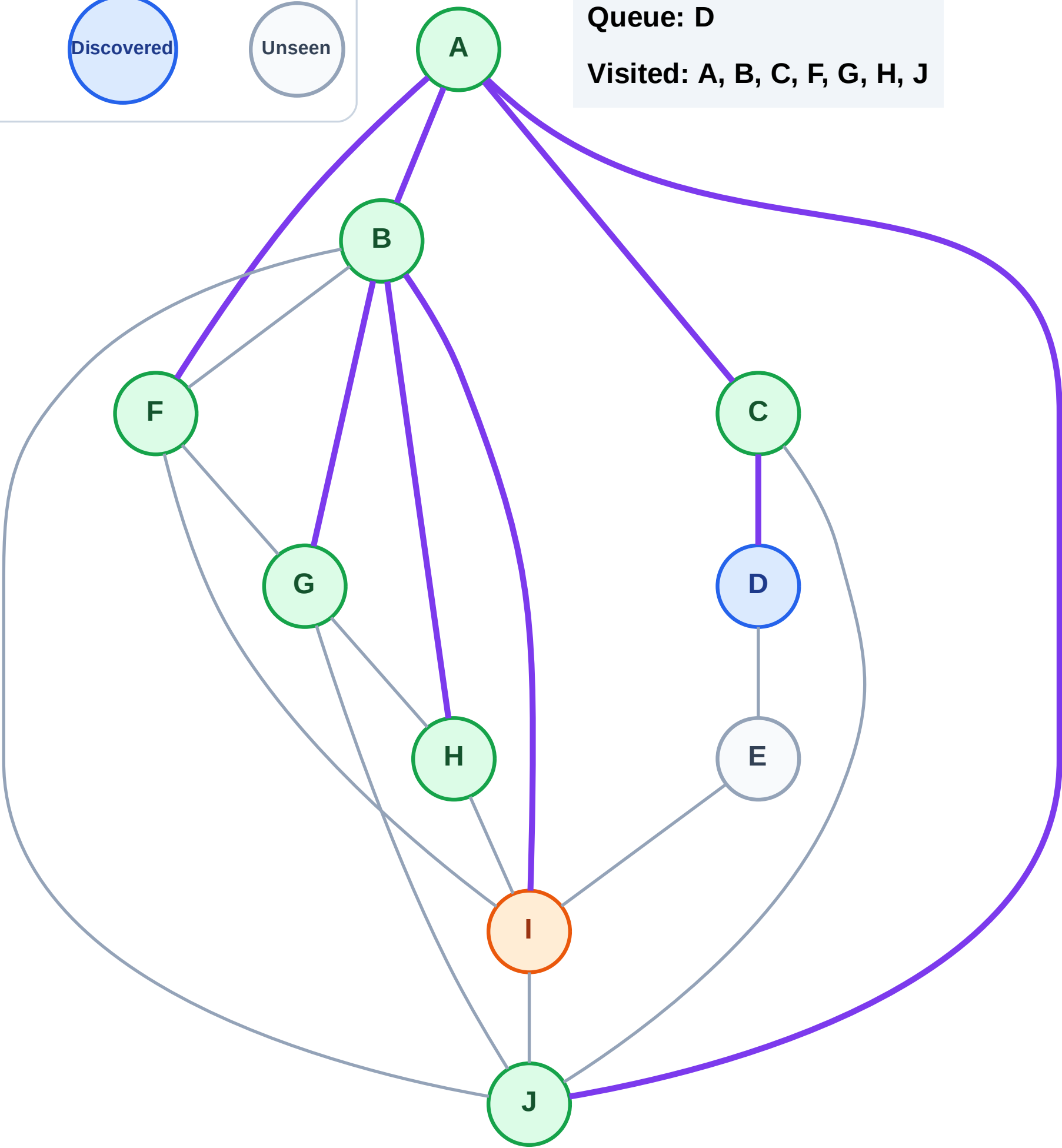
Discovered



Unseen

Queue: D

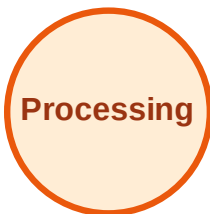
Visited: A, B, C, F, G, H, J



BFS Traversal

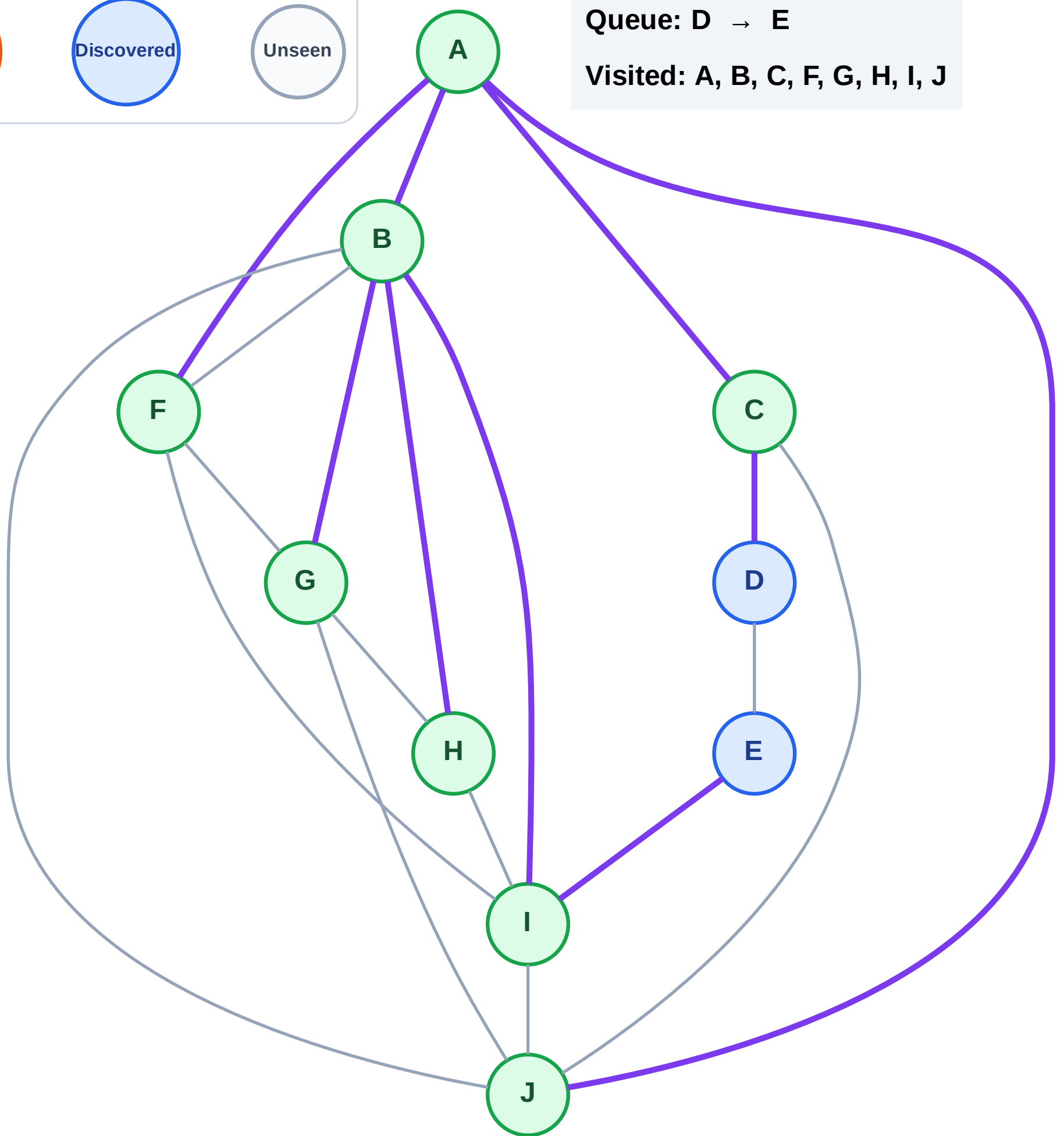
Step 17: Discover E from I

Legend



Queue: D → E

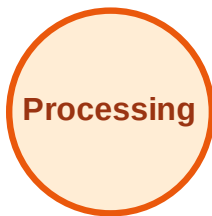
Visited: A, B, C, F, G, H, I, J



BFS Traversal

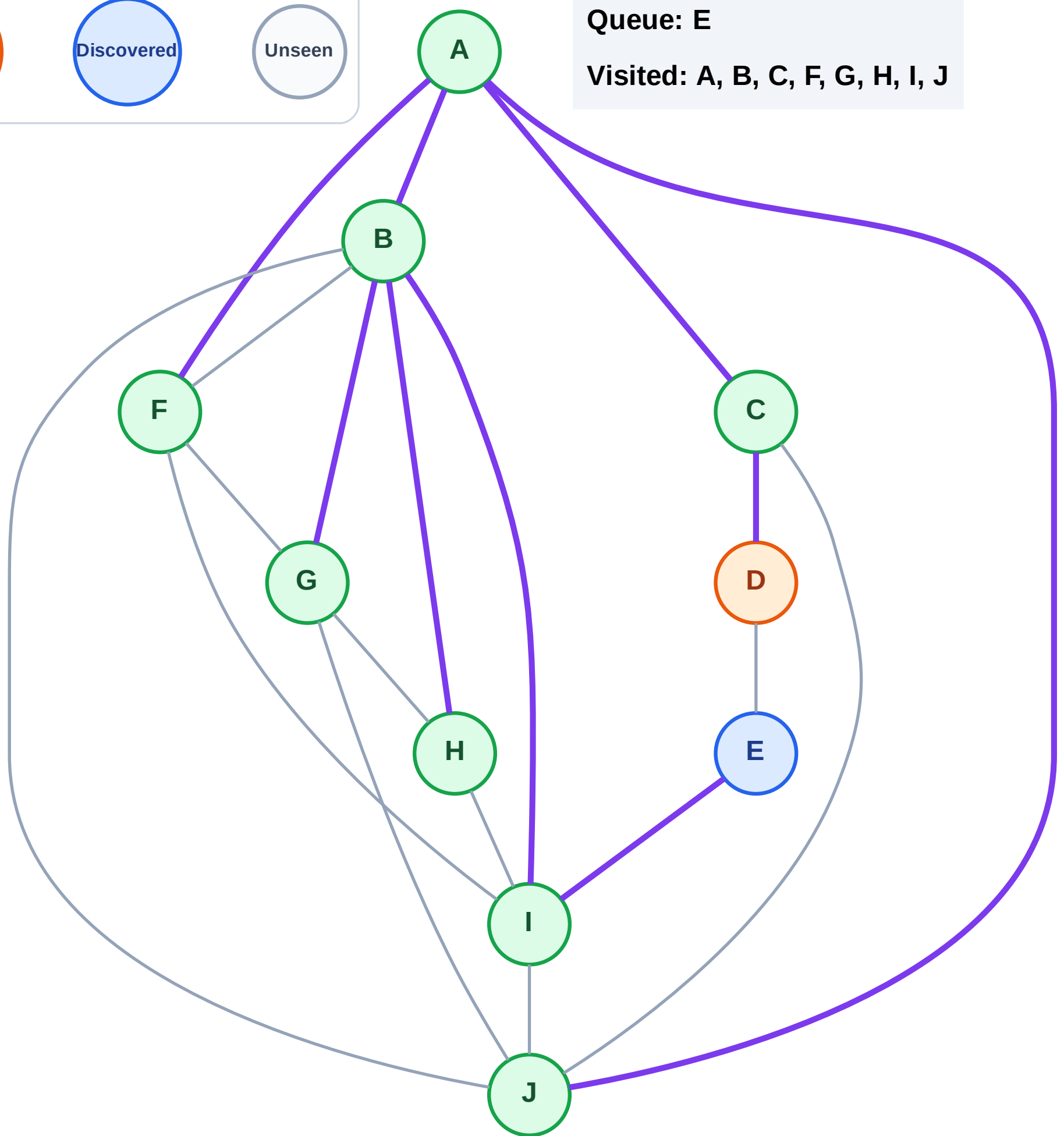
Step 18: Process node D

Legend



Queue: E

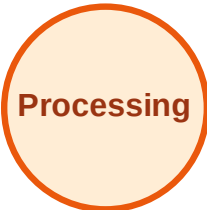
Visited: A, B, C, F, G, H, I, J



BFS Traversal

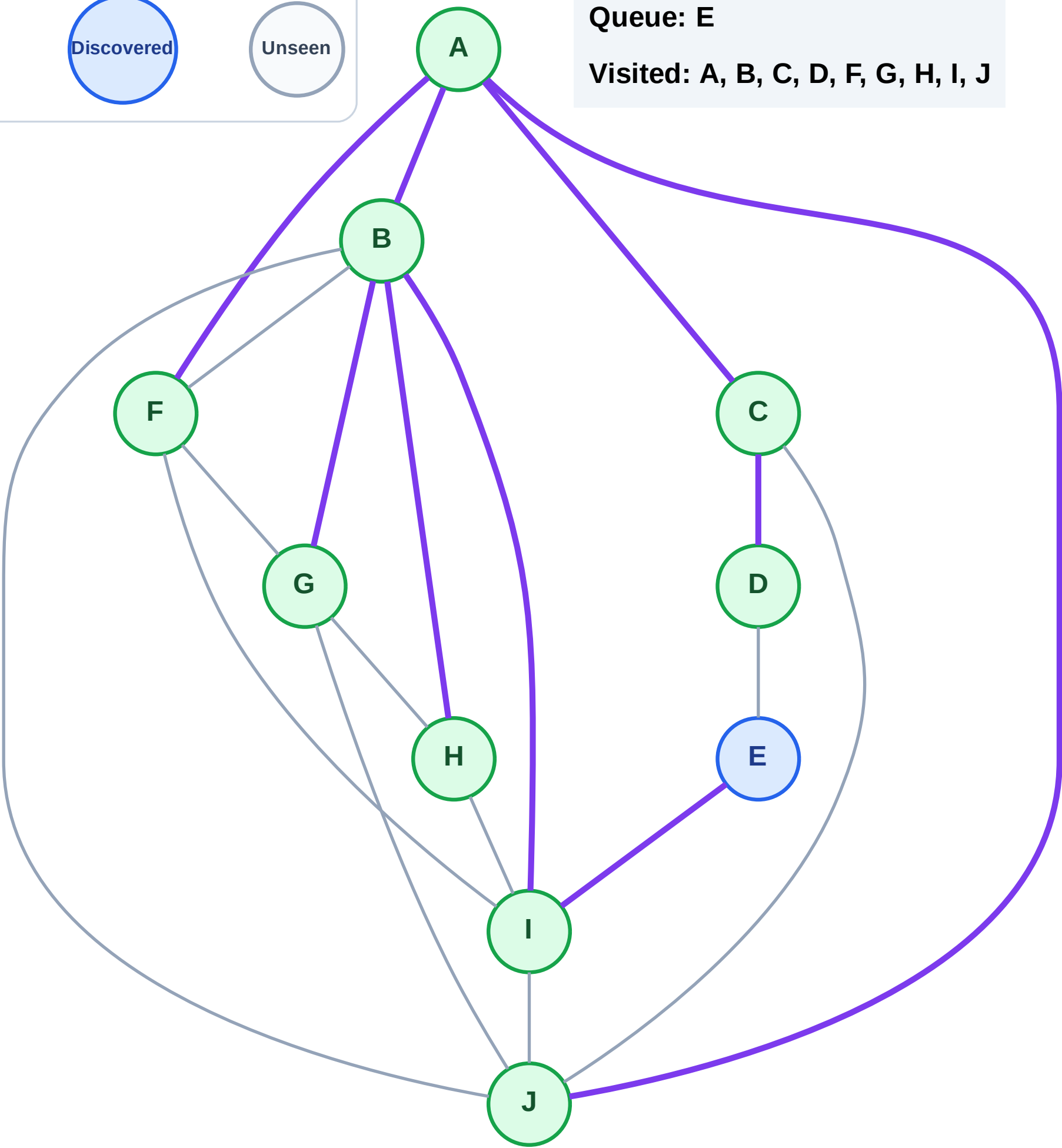
Step 19: No new neighbors from D

Legend



Queue: E

Visited: A, B, C, D, F, G, H, I, J



BFS Traversal

Step 20: Process node E

Legend



Complete



Processing



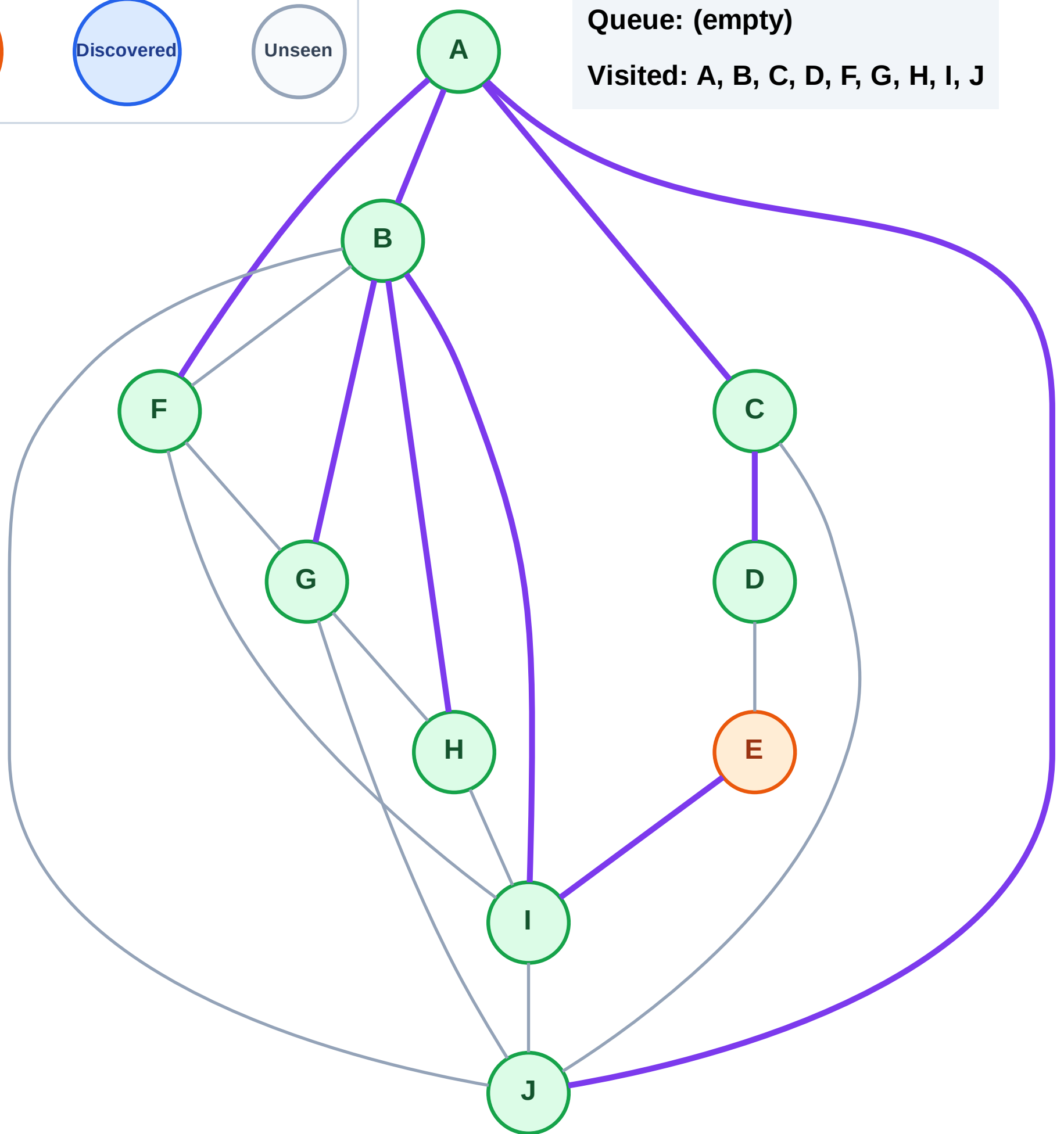
Discovered



Unseen

Queue: (empty)

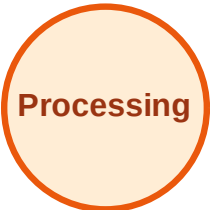
Visited: A, B, C, D, F, G, H, I, J



BFS Traversal

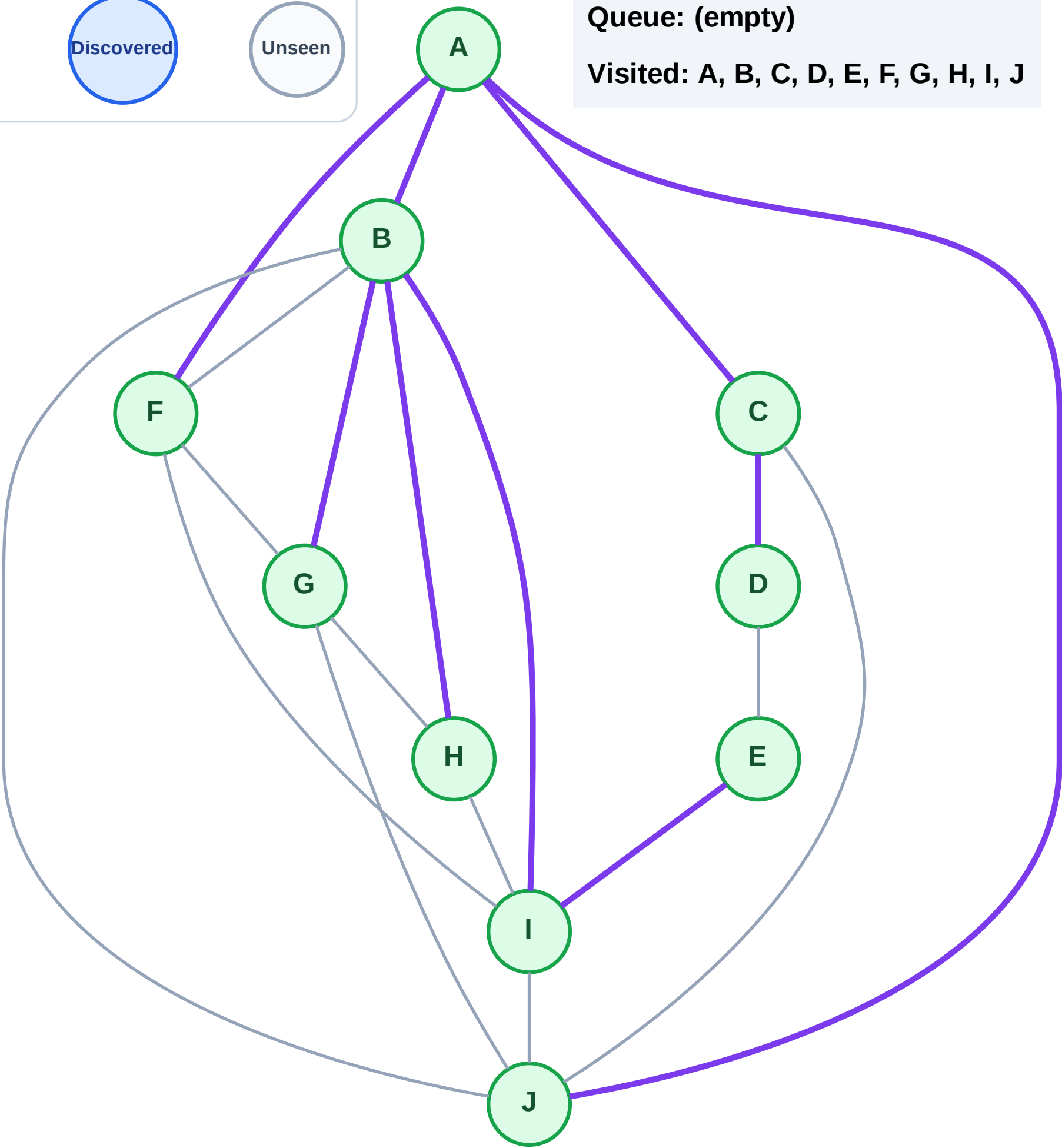
Step 21: No new neighbors from E

Legend



Queue: (empty)

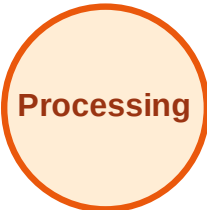
Visited: A, B, C, D, E, F, G, H, I, J



DFS Traversal

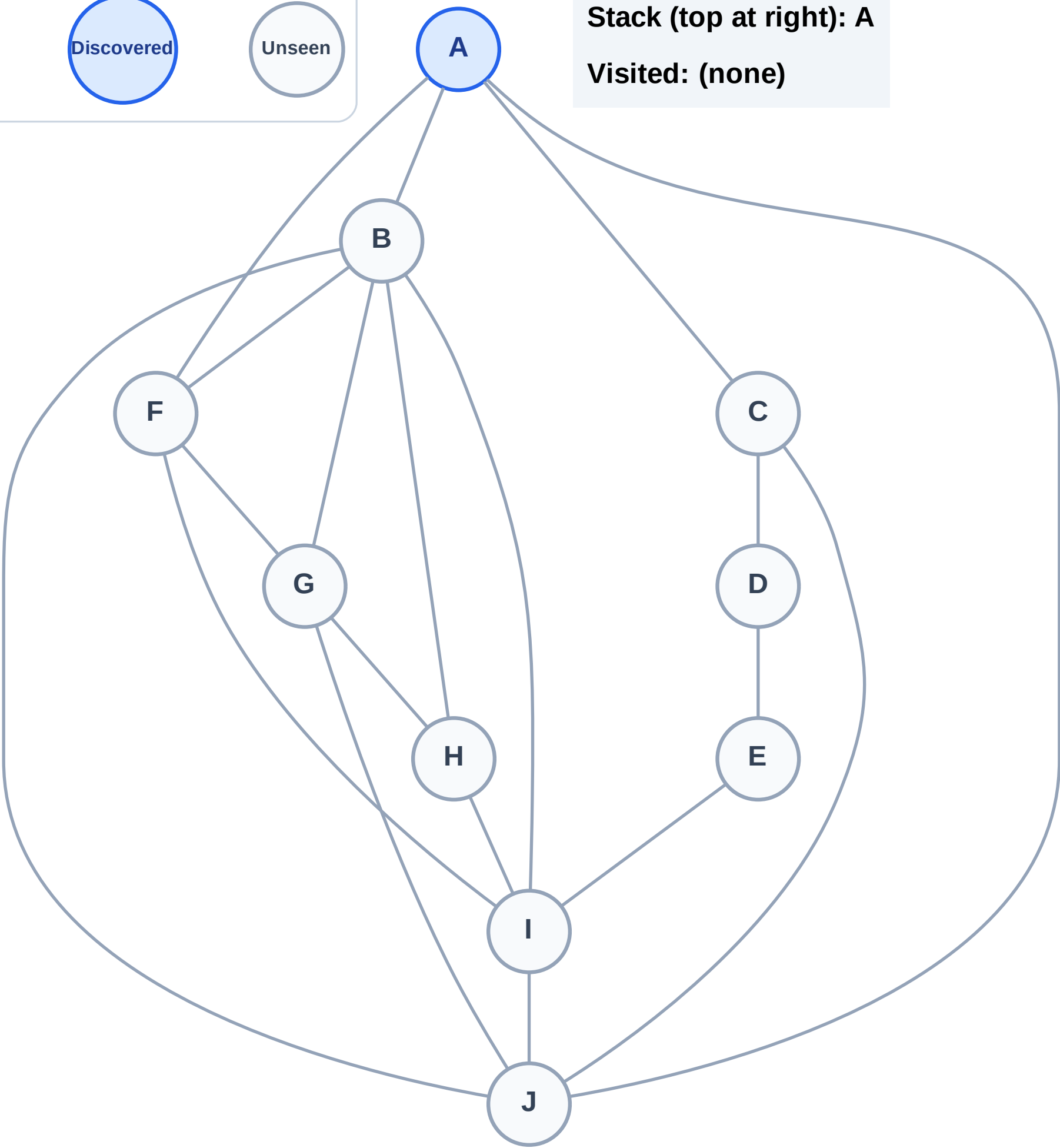
Step 1: Start at A

Legend



Stack (top at right): A

Visited: (none)



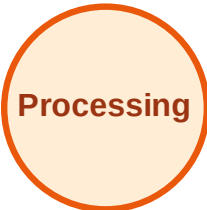
DFS Traversal

Step 2: Process node A

Legend



Processing



Discovered

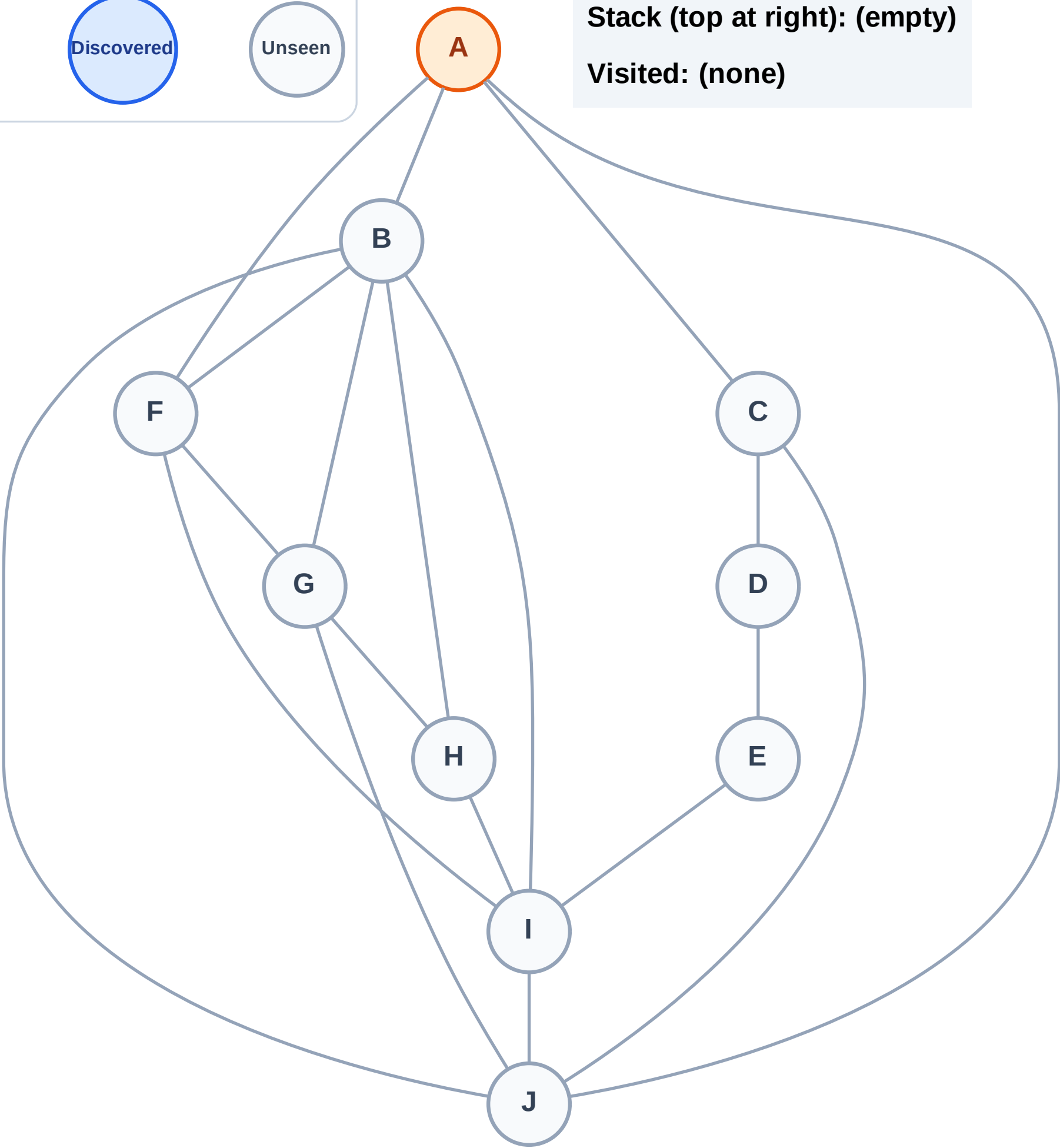


Unseen



Stack (top at right): (empty)

Visited: (none)



DFS Traversal

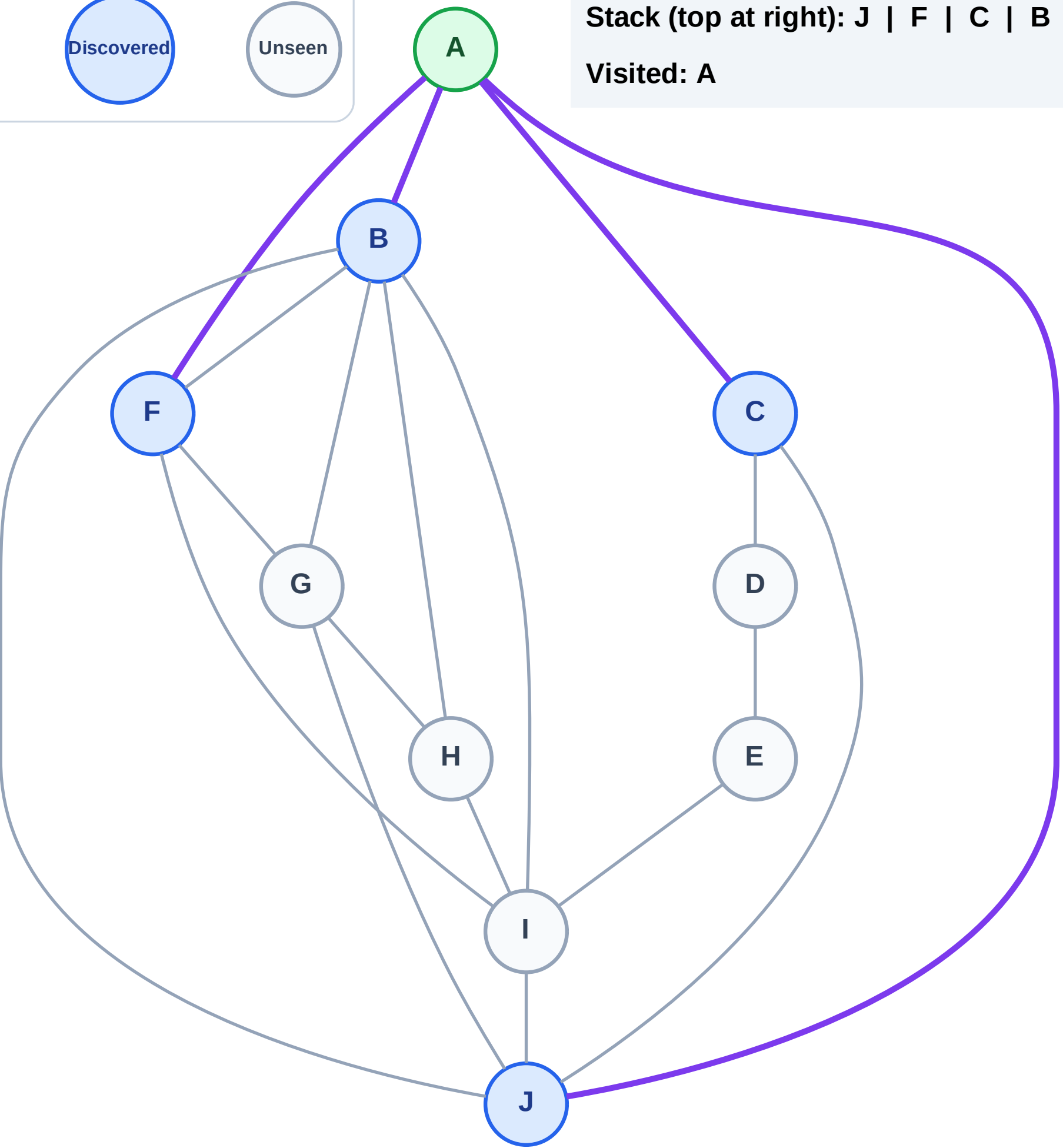
Step 3: Discover J, F, C, B from A

Legend



Stack (top at right): J | F | C | B

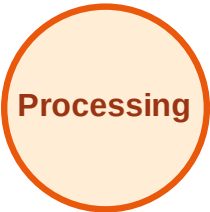
Visited: A



DFS Traversal

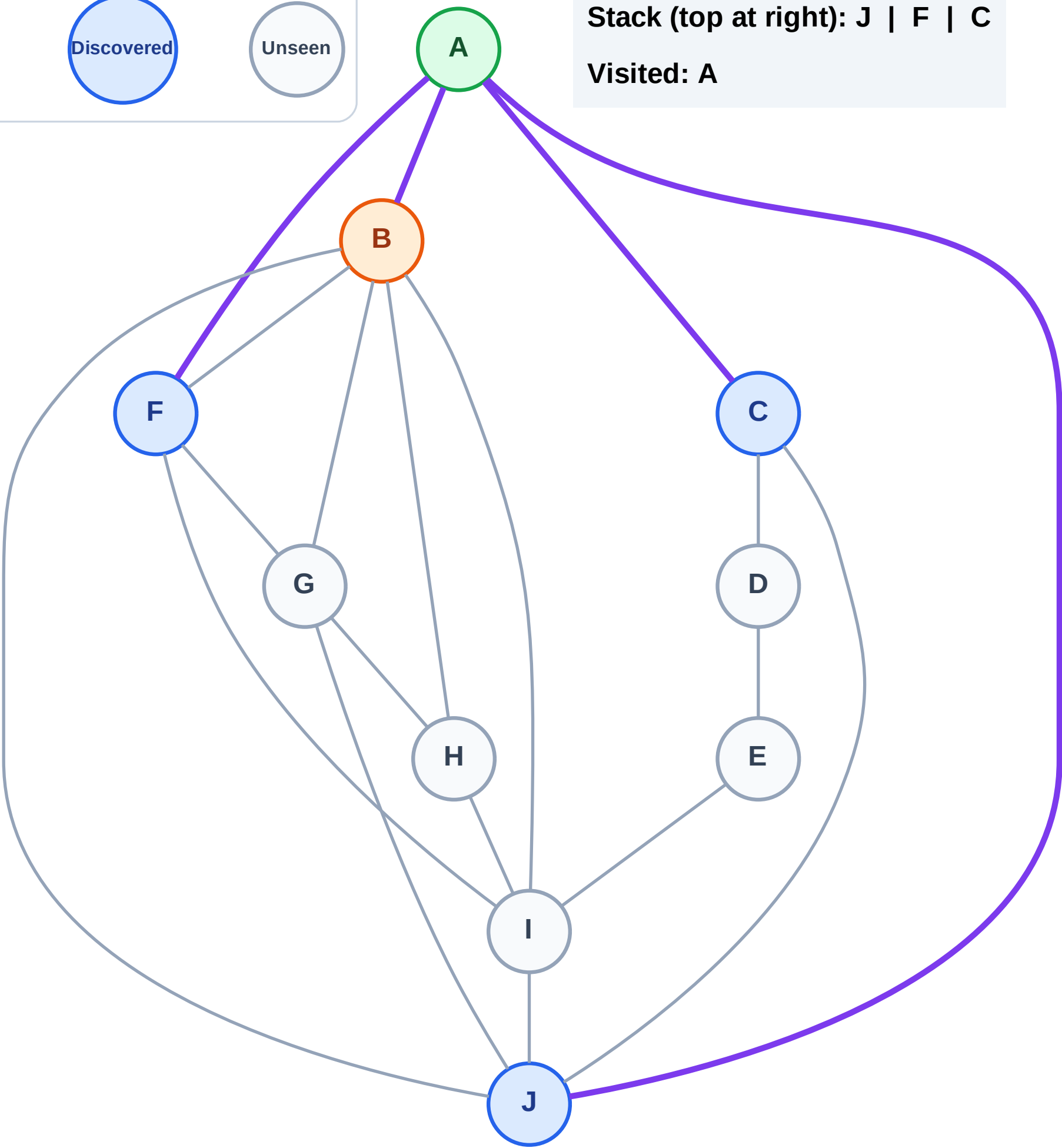
Step 4: Process node B

Legend



Stack (top at right): J | F | C

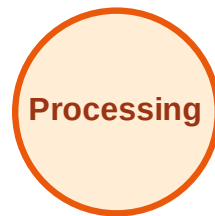
Visited: A



DFS Traversal

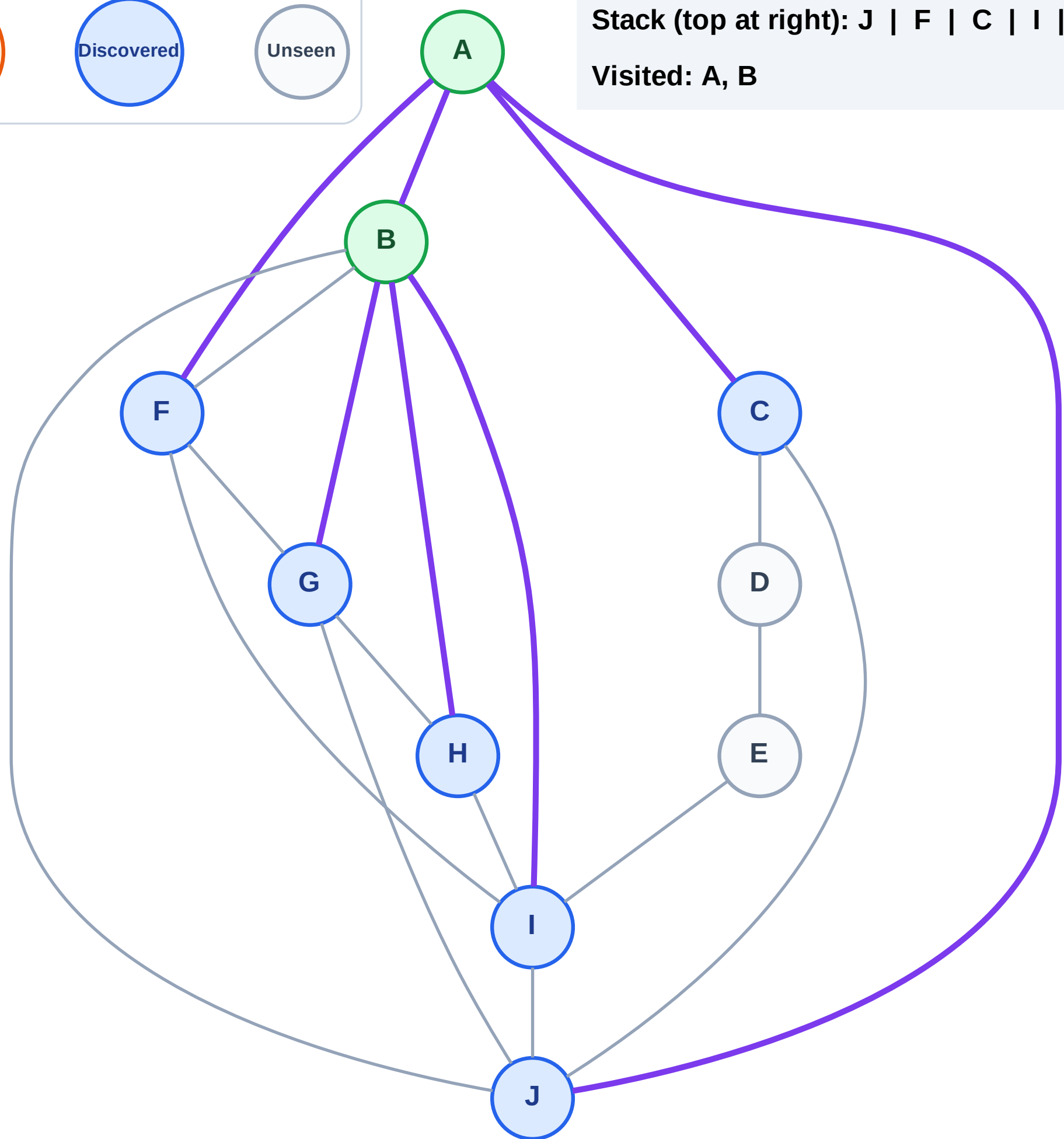
Step 5: Discover I, H, G from B

Legend



Stack (top at right): J | F | C | I | H | G

Visited: A, B



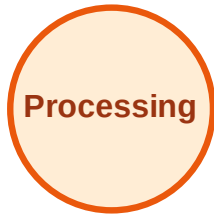
DFS Traversal

Step 6: Process node G

Legend



Complete



Processing



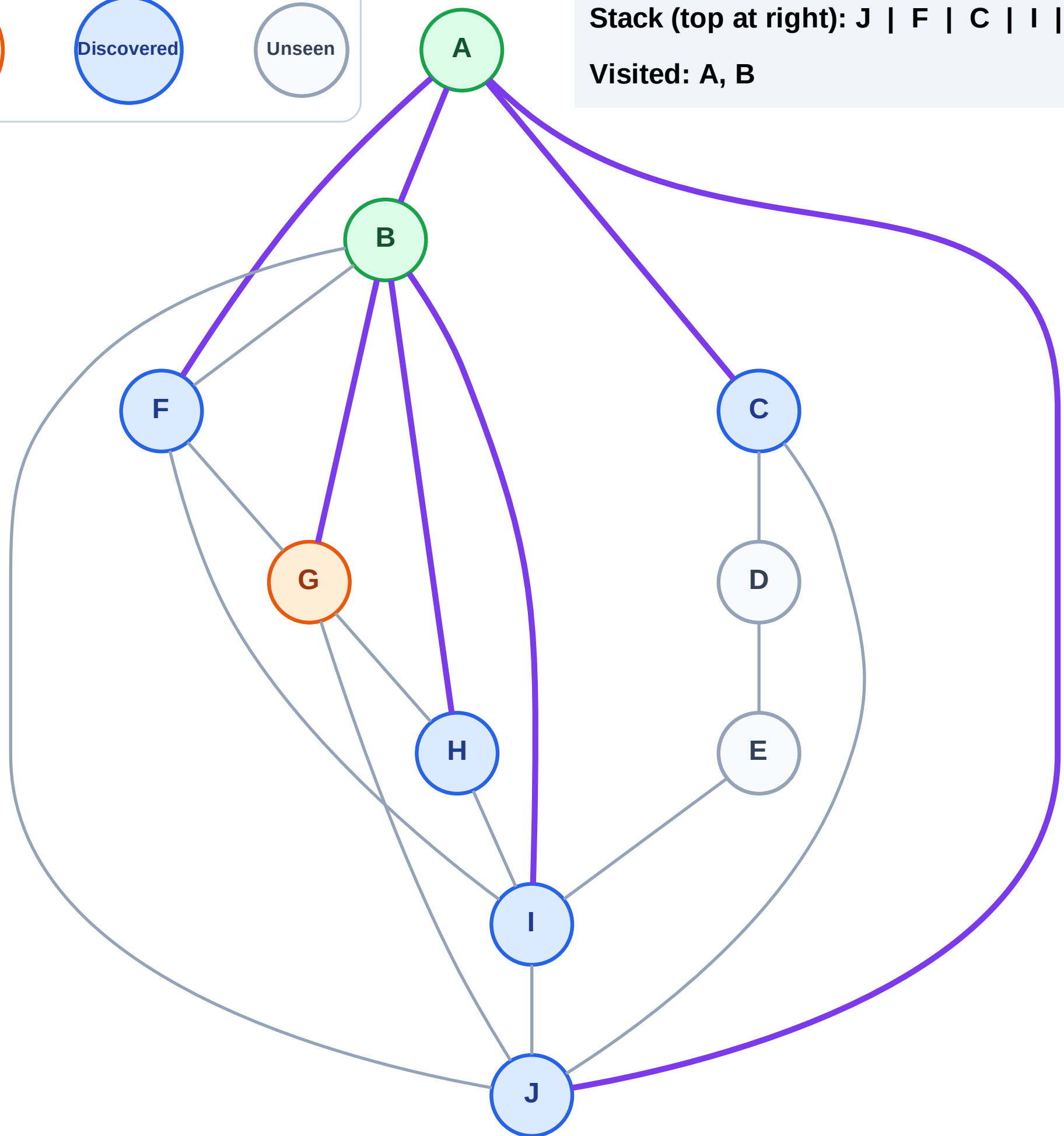
Discovered



Unseen

Stack (top at right): J | F | C | I | H

Visited: A, B



DFS Traversal

Step 7: No new neighbors from G

Legend



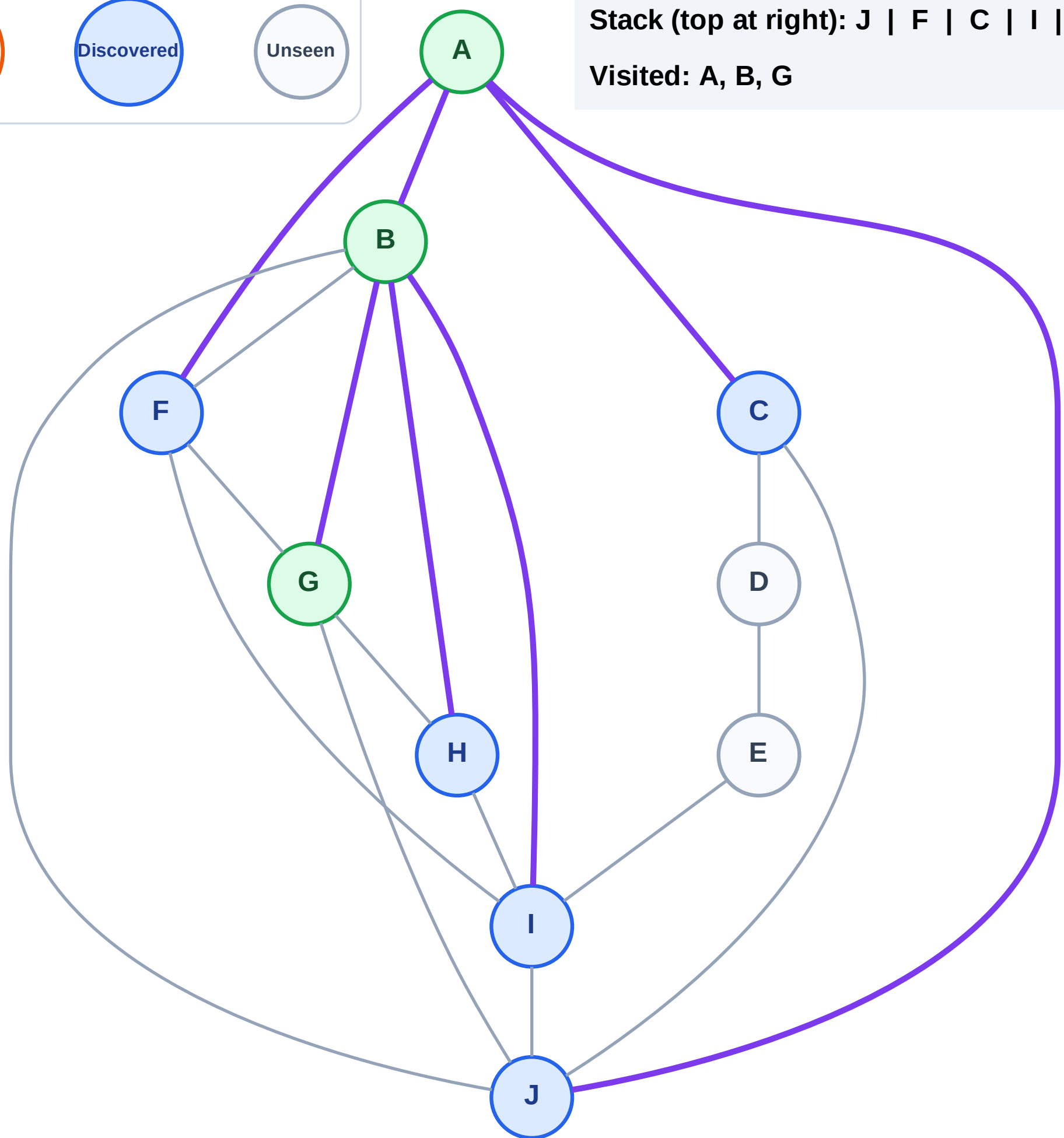
Processing



Unseen

Stack (top at right): J | F | C | I | H

Visited: A, B, G



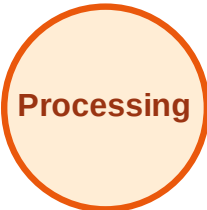
DFS Traversal

Step 8: Process node H

Legend



Complete



Processing



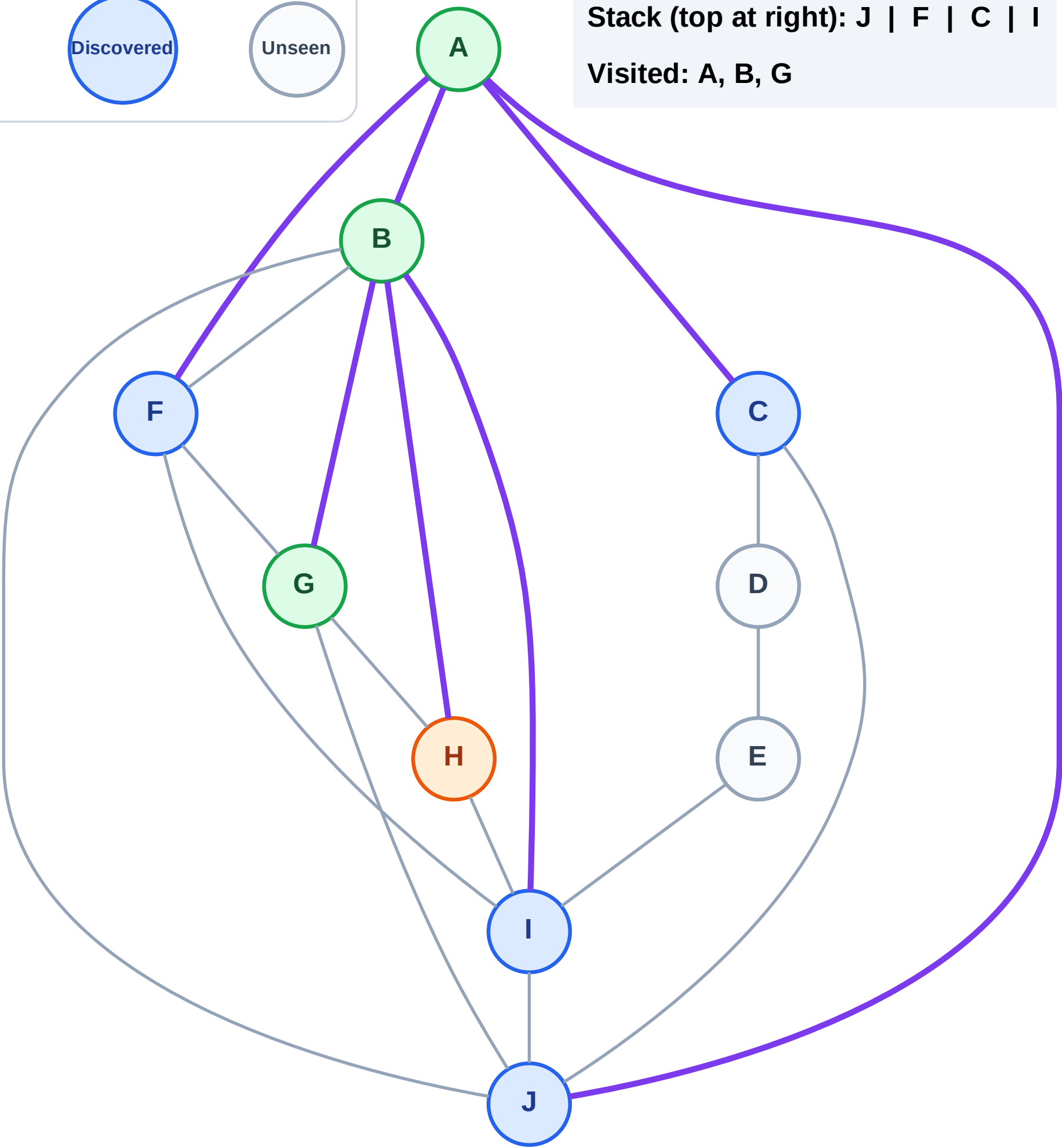
Discovered



Unseen

Stack (top at right): J | F | C | I

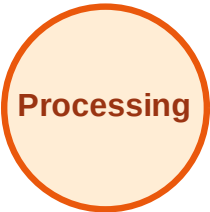
Visited: A, B, G



DFS Traversal

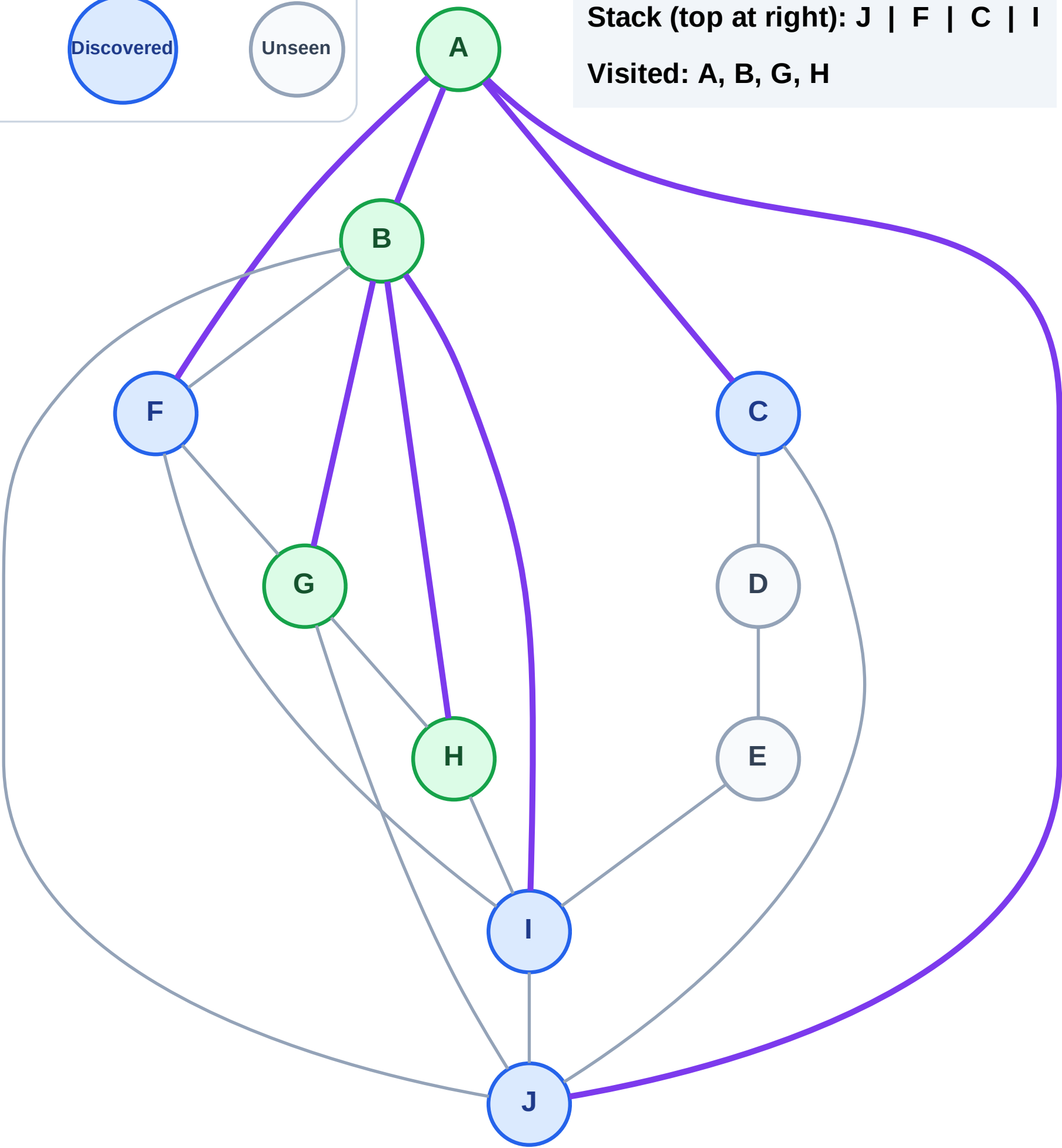
Step 9: No new neighbors from H

Legend



Stack (top at right): J | F | C | I

Visited: A, B, G, H



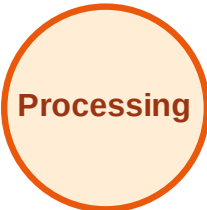
DFS Traversal

Step 10: Process node I

Legend



Complete



Processing



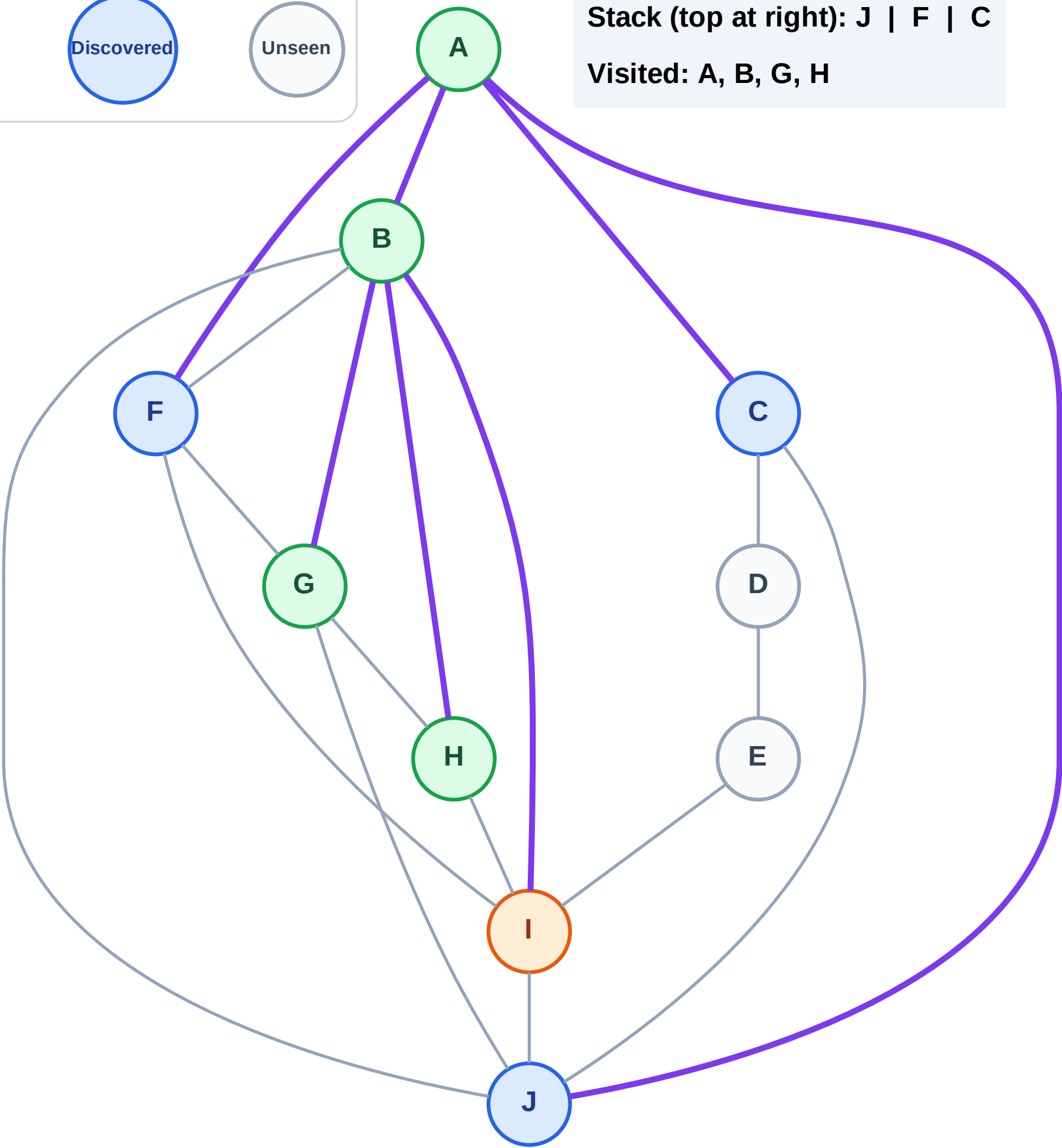
Discovered



Unseen

Stack (top at right): J | F | C

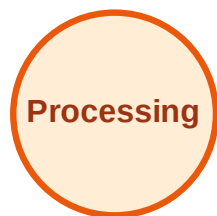
Visited: A, B, G, H



DFS Traversal

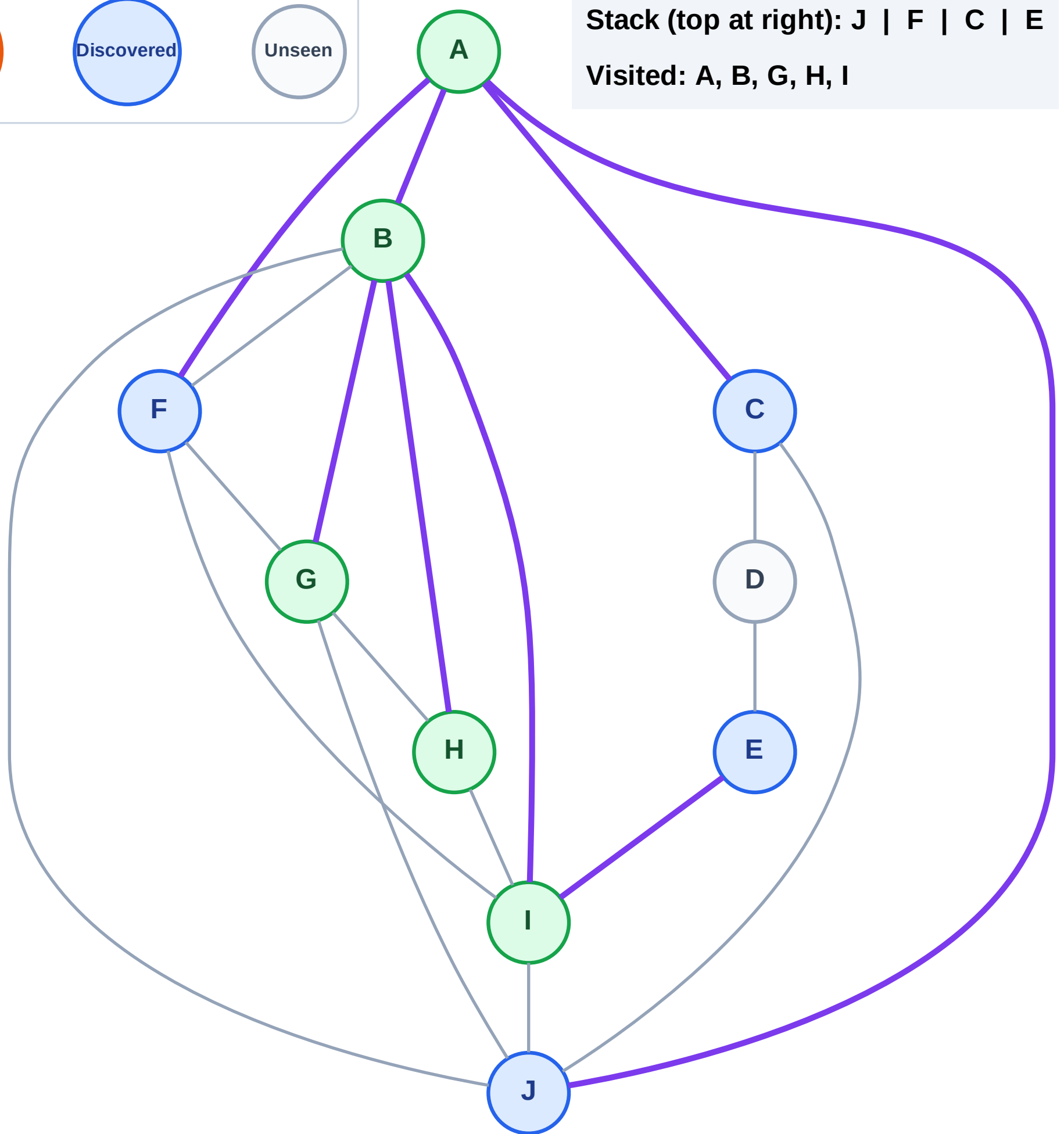
Step 11: Discover E from I

Legend



Stack (top at right): J | F | C | E

Visited: A, B, G, H, I



DFS Traversal

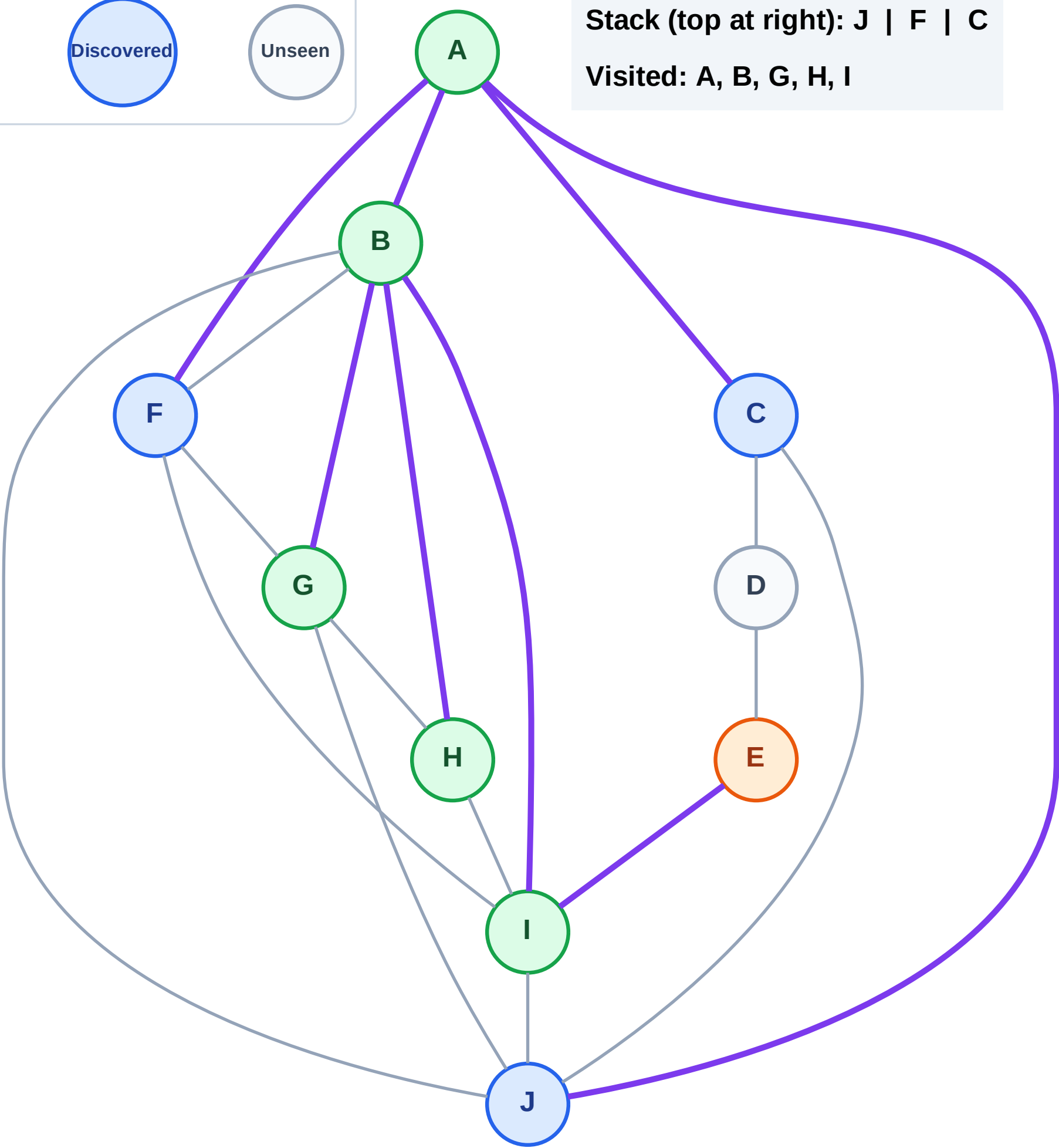
Step 12: Process node E

Legend



Stack (top at right): J | F | C

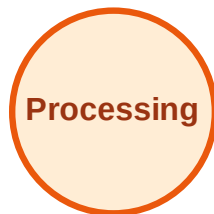
Visited: A, B, G, H, I



DFS Traversal

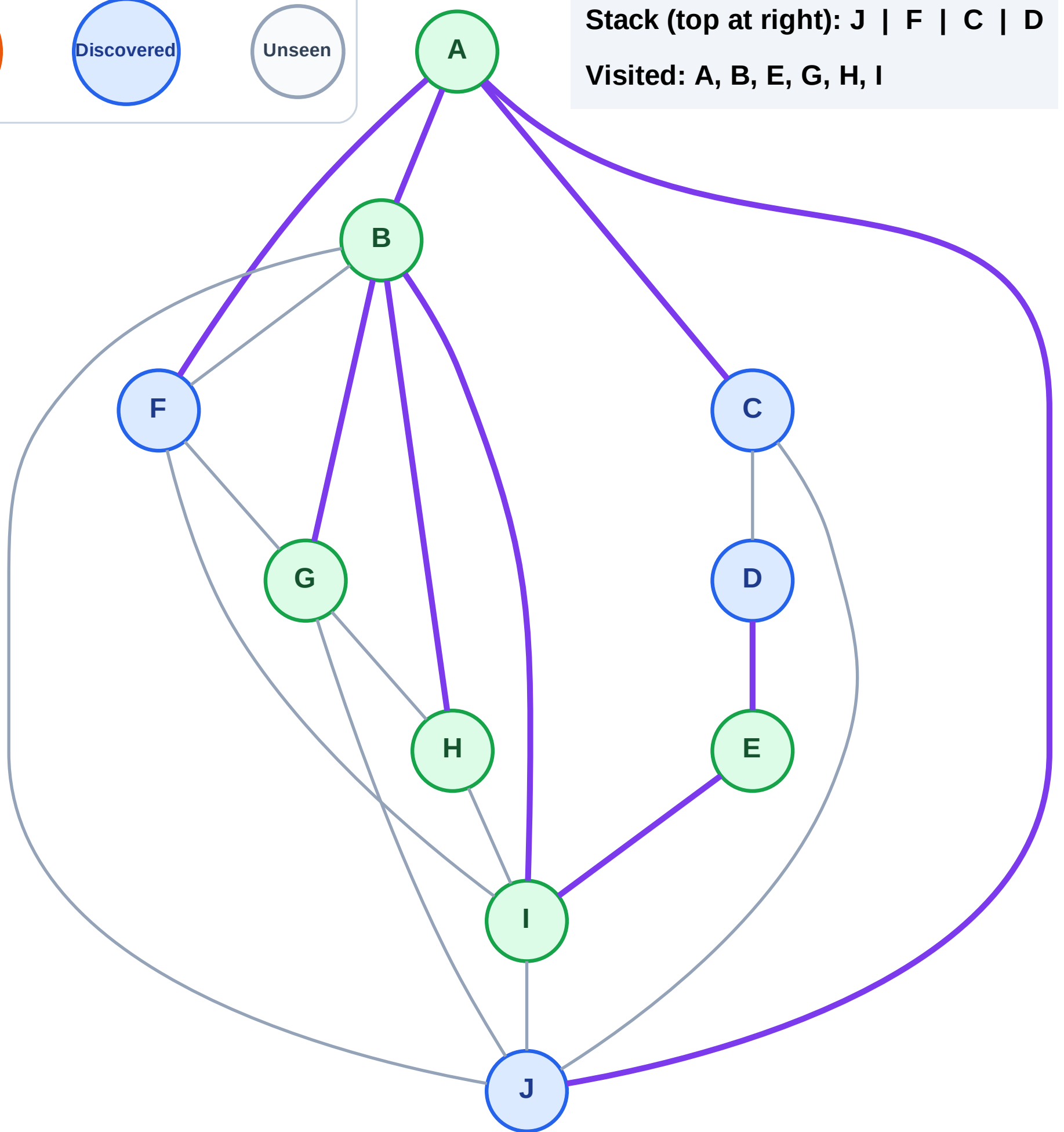
Step 13: Discover D from E

Legend



Stack (top at right): J | F | C | D

Visited: A, B, E, G, H, I



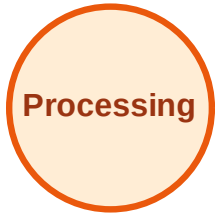
DFS Traversal

Step 14: Process node D

Legend



Complete



Processing



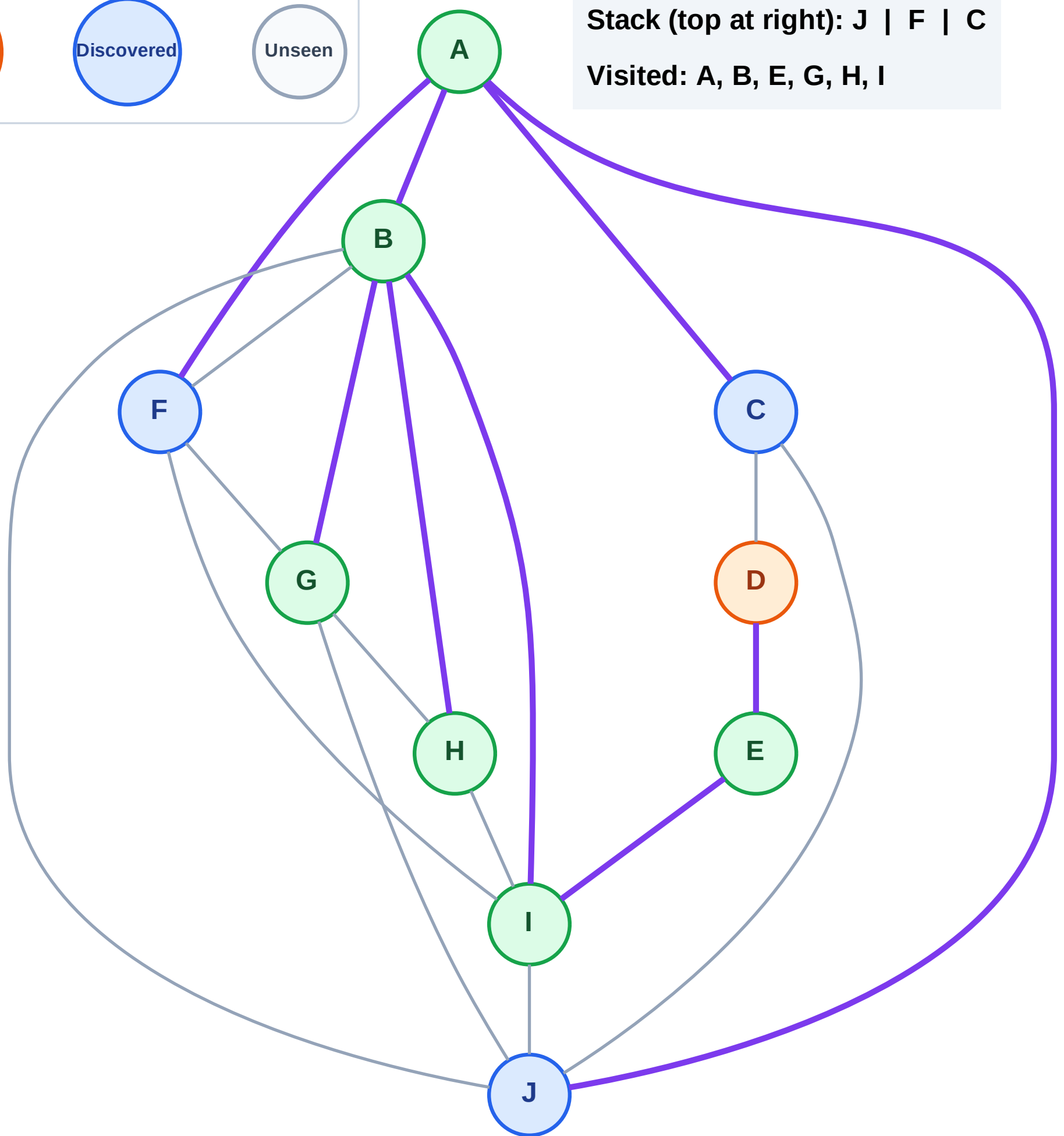
Discovered



Unseen

Stack (top at right): J | F | C

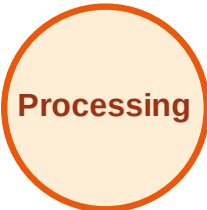
Visited: A, B, E, G, H, I



DFS Traversal

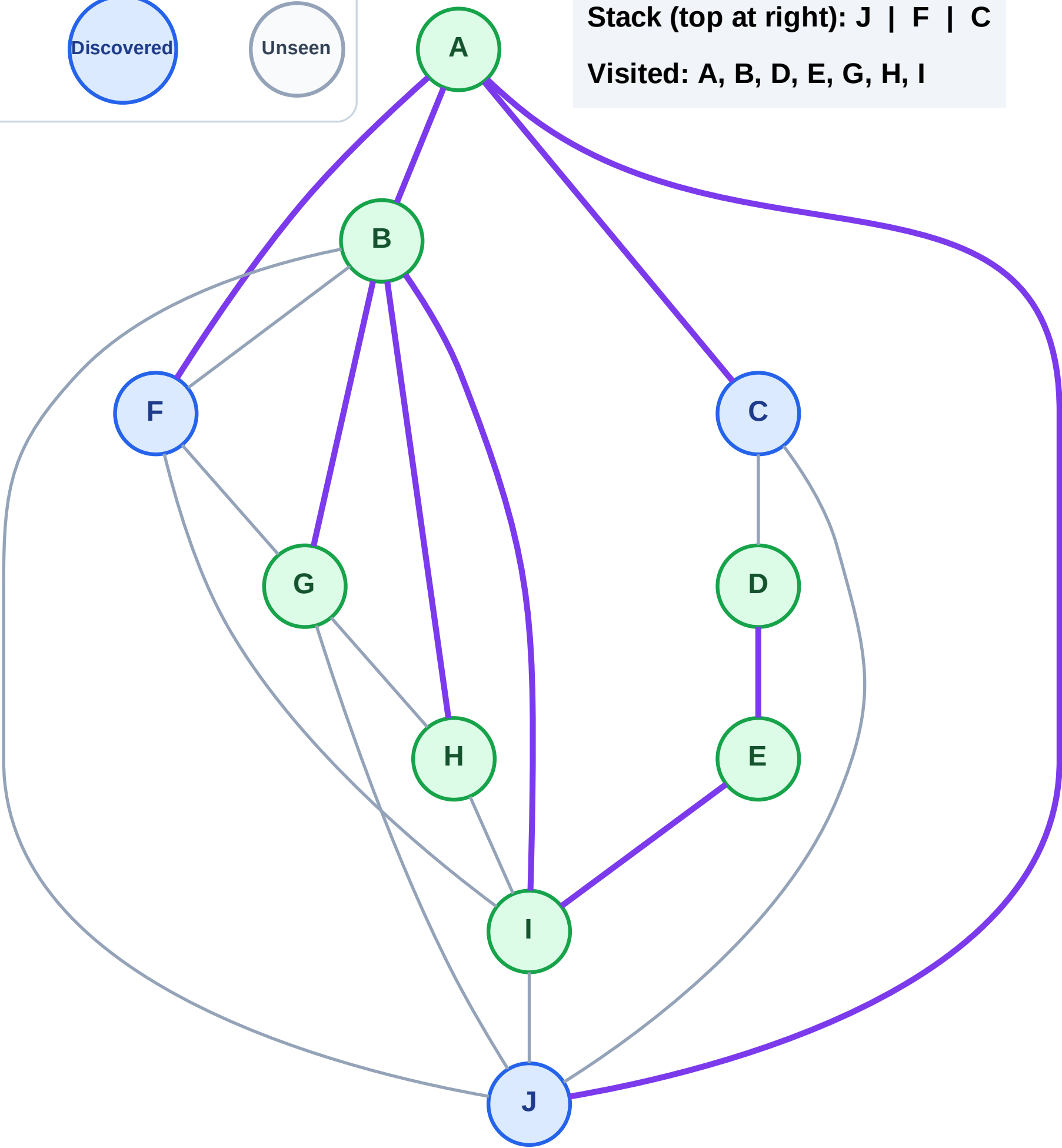
Step 15: No new neighbors from D

Legend



Stack (top at right): J | F | C

Visited: A, B, D, E, G, H, I



DFS Traversal

Step 16: Process node C

Legend



Complete



Processing



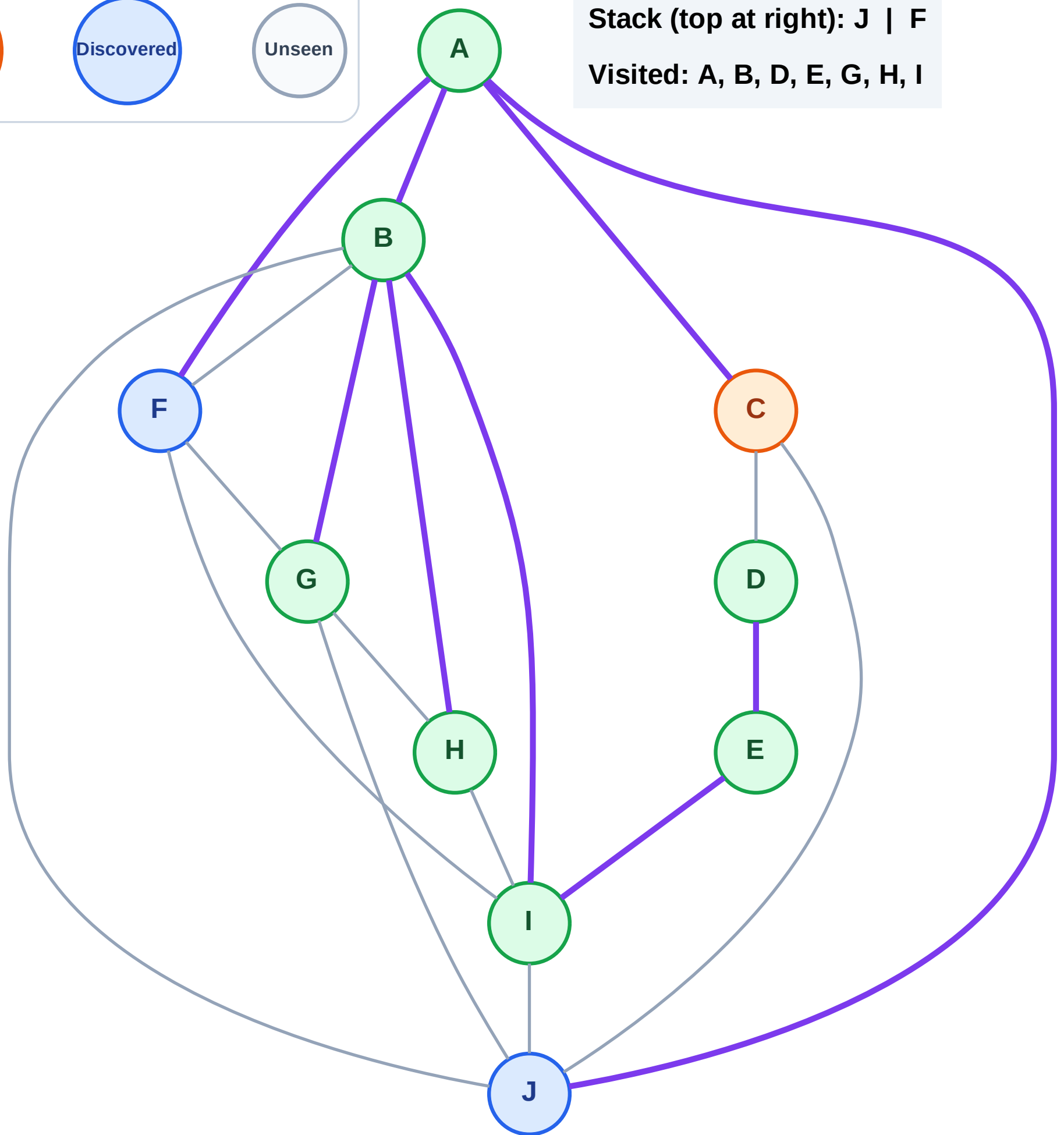
Discovered



Unseen

Stack (top at right): J | F

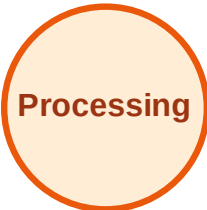
Visited: A, B, D, E, G, H, I



DFS Traversal

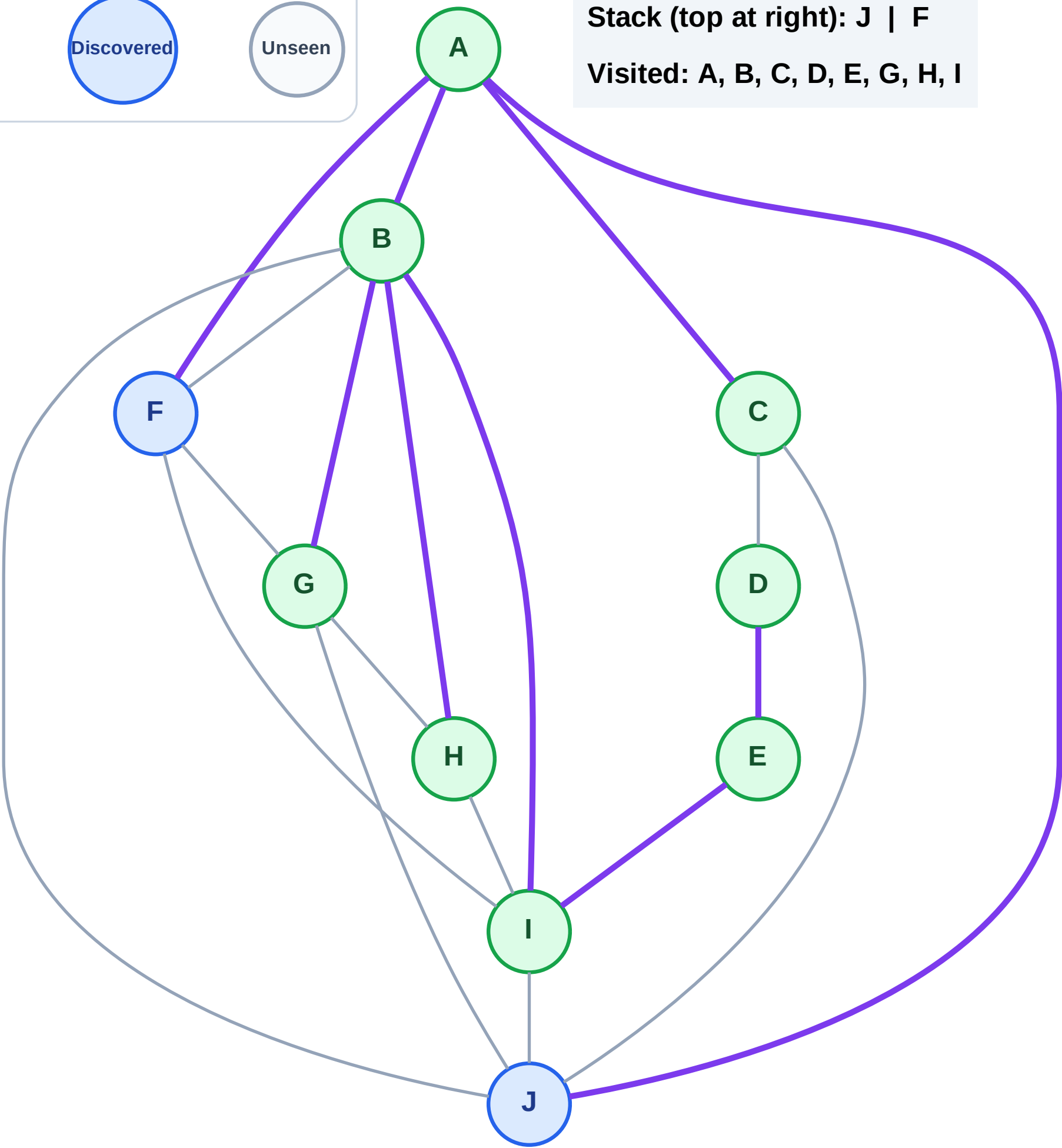
Step 17: No new neighbors from C

Legend



Stack (top at right): J | F

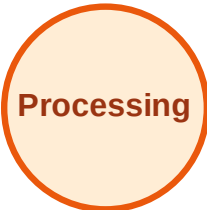
Visited: A, B, C, D, E, G, H, I



DFS Traversal

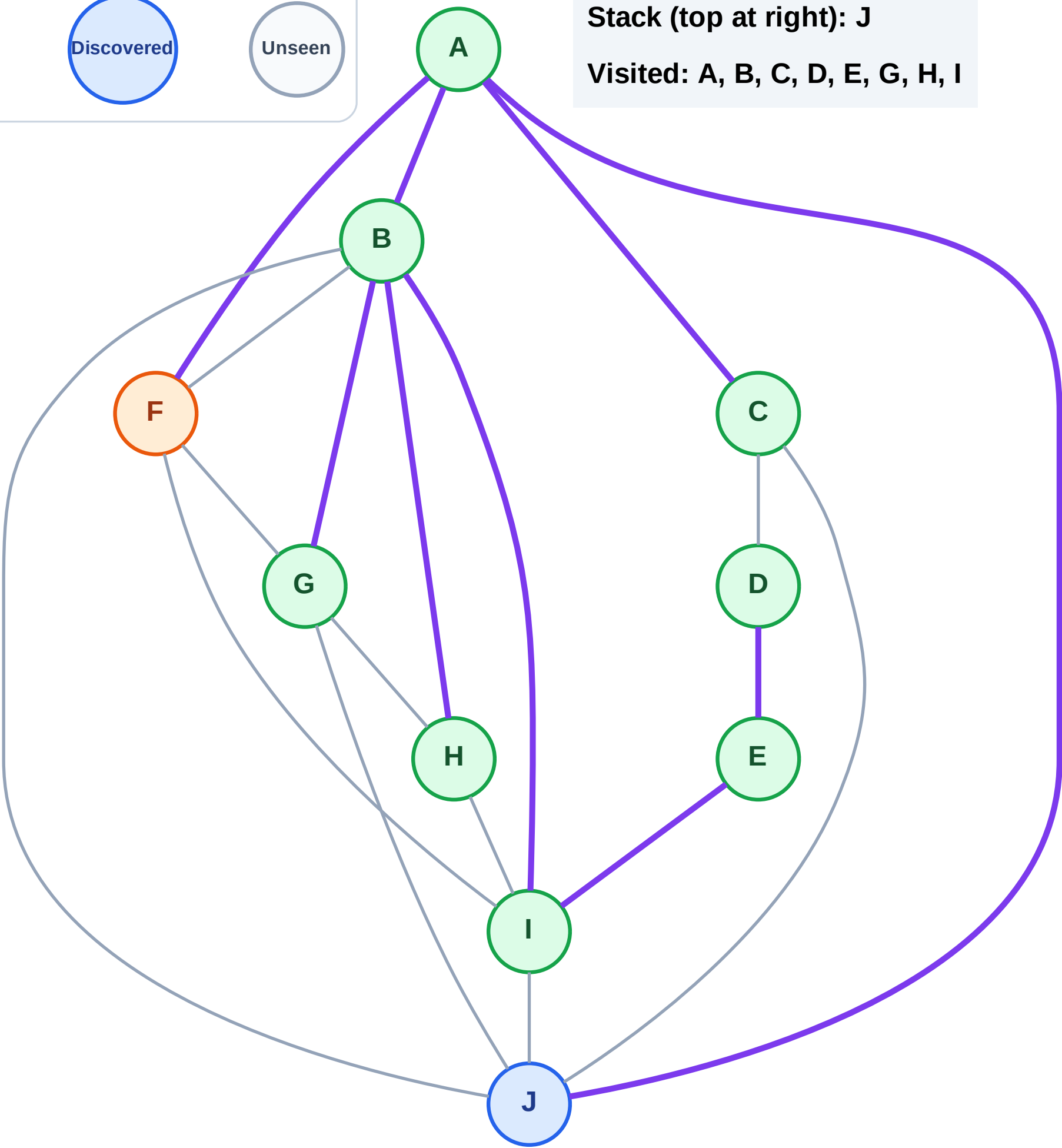
Step 18: Process node F

Legend



Stack (top at right): J

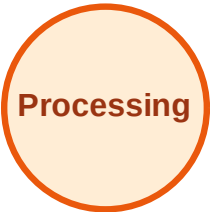
Visited: A, B, C, D, E, G, H, I



DFS Traversal

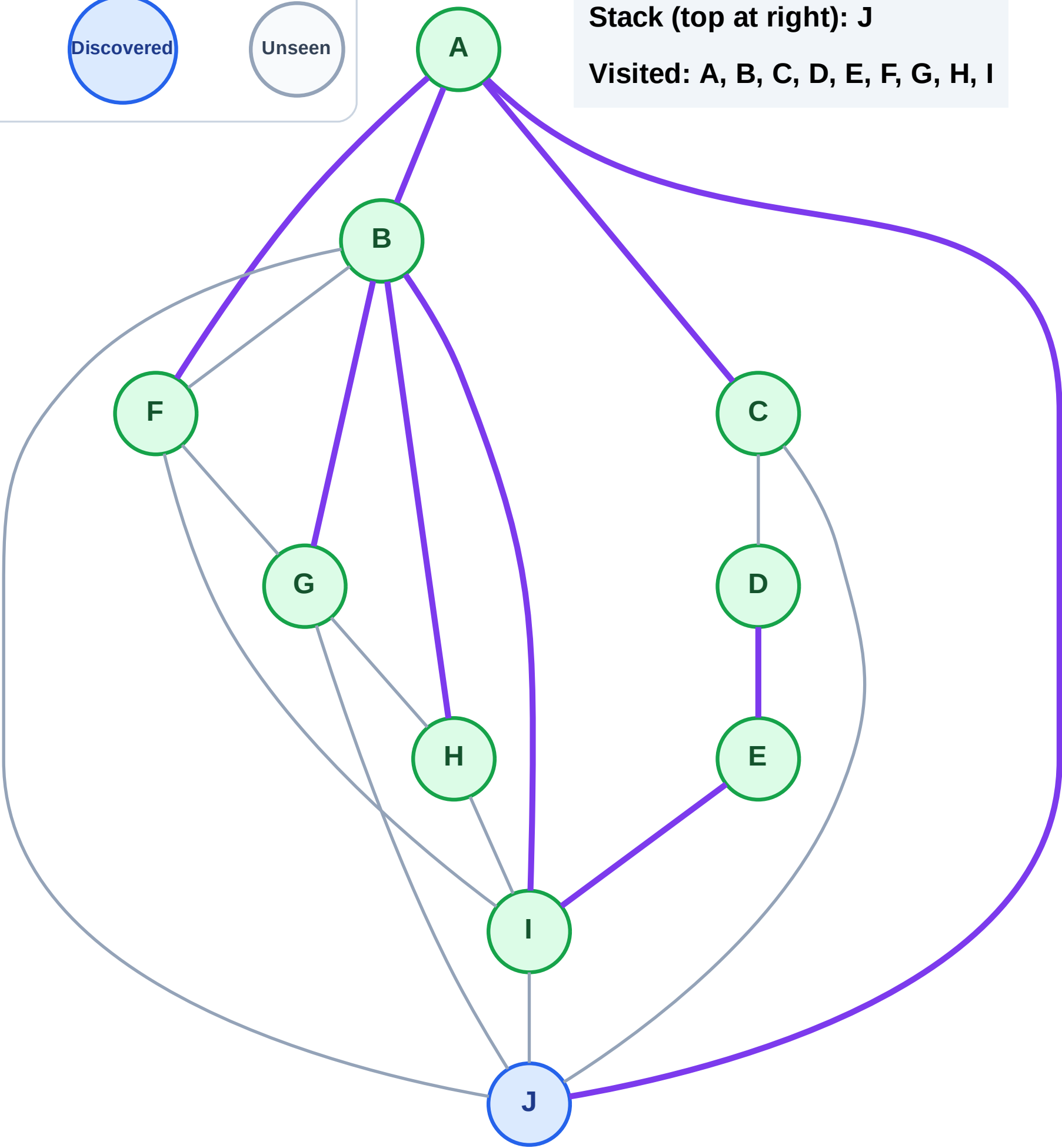
Step 19: No new neighbors from F

Legend



Stack (top at right): J

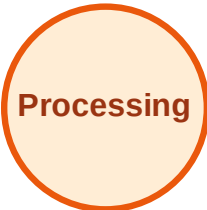
Visited: A, B, C, D, E, F, G, H, I



DFS Traversal

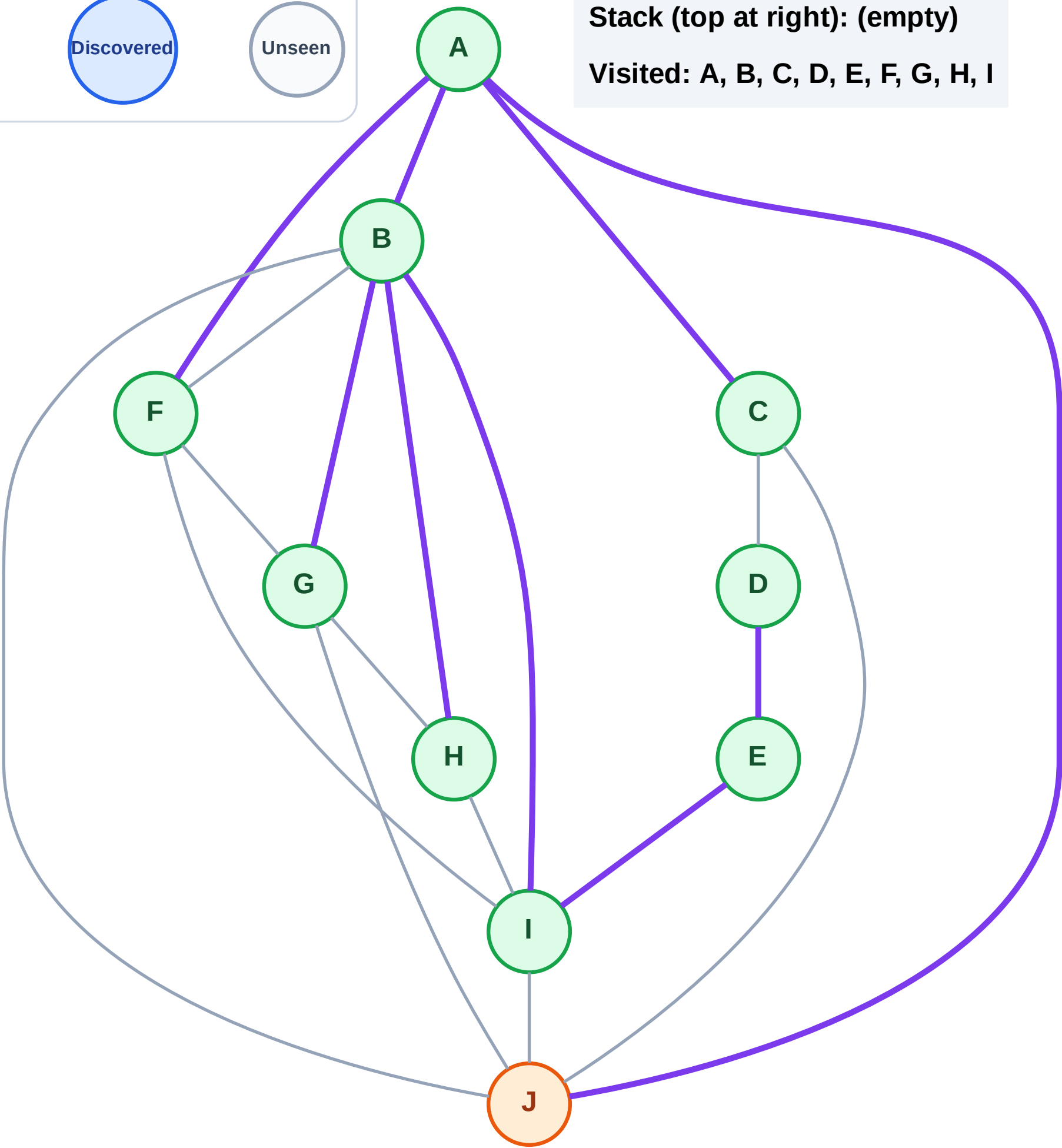
Step 20: Process node J

Legend



Stack (top at right): (empty)

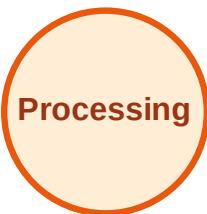
Visited: A, B, C, D, E, F, G, H, I



DFS Traversal

Step 21: No new neighbors from J

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

